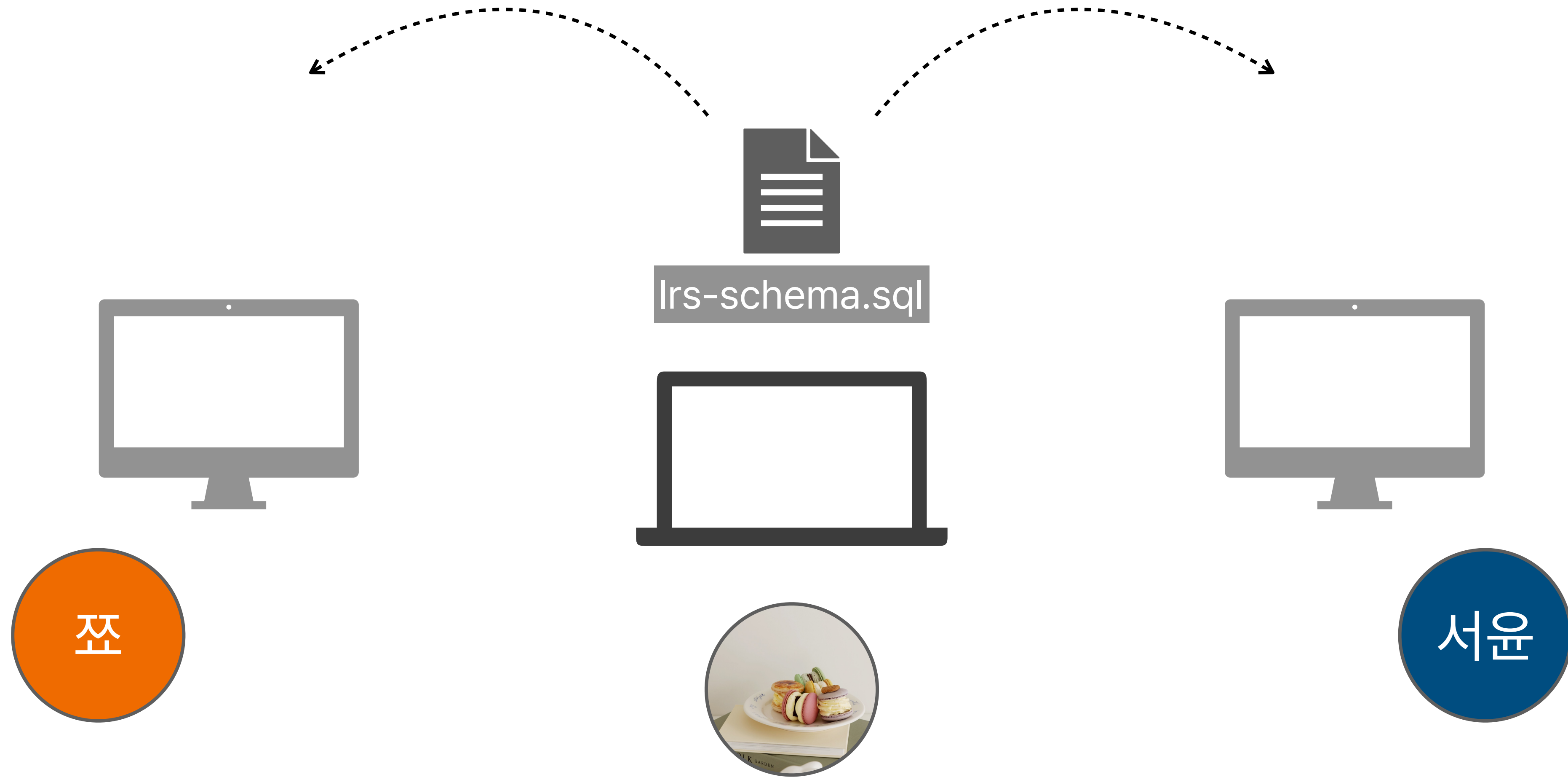


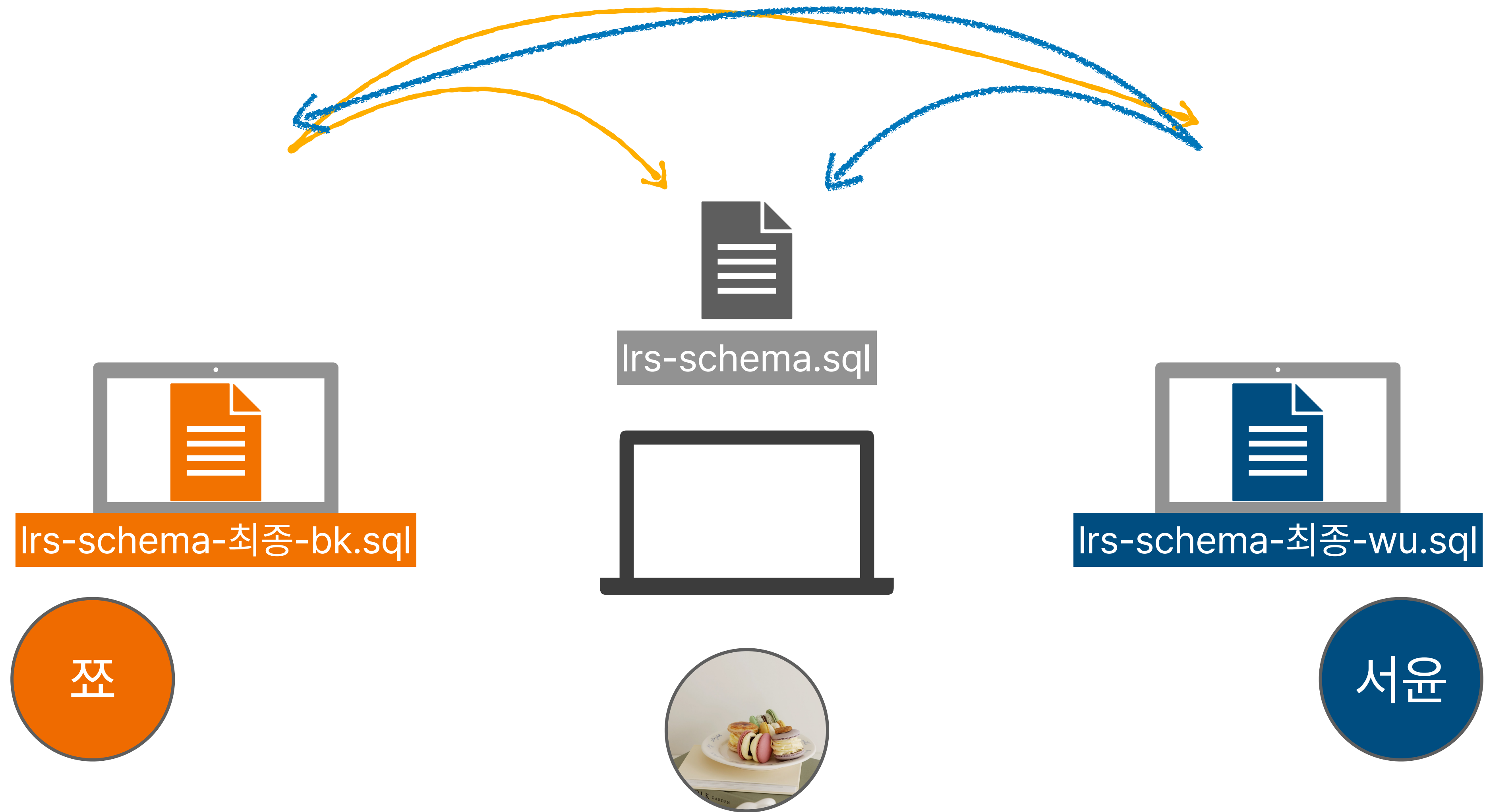
Git & Github Guide for Forest🌳

🌊 Vibe Coder🏄 관점에서...강의분석팀의 원활한 LRS 개발을 위한...

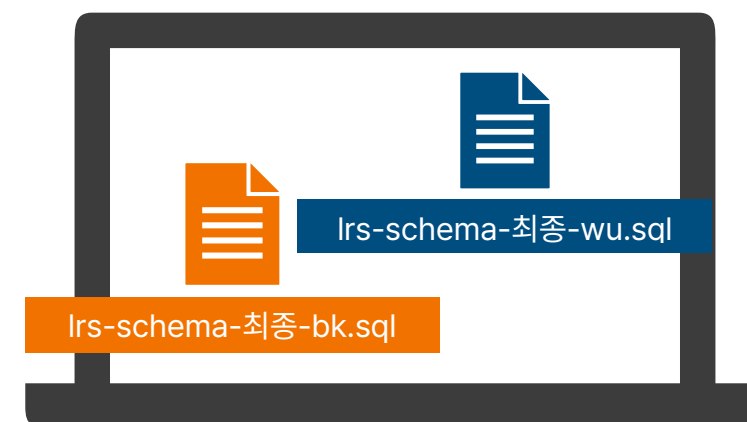
Maybe..



Maybe..



Maybe..



Maybe..

ctrl + c ?

ctrl + v ?

```
CREATE TABLE IF NOT EXISTS applications (  
  id UUID PRIMARY KEY DEFAULT  
  gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE  
  CASCADE,  
  week_number INTEGER NOT NULL,  
  subject_id UUID REFERENCES subjects(id) ON  
  DELETE SET NULL,  
  requested_count INTEGER NOT NULL CHECK  
  (requested_count BETWEEN 1 AND 3),  
  status application_status DEFAULT 'pending',  
  created_at TIMESTAMPTZ DEFAULT NOW(),  
  object_id  
)
```



lrs-schema-최종-bk.sql

```
CREATE TABLE IF NOT EXISTS applications (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE  
  CASCADE,  
  week_number INTEGER NOT NULL,  
  requested_count INTEGER NOT NULL CHECK  
  (requested_count BETWEEN 1 AND 3),  
  status application_status DEFAULT 'pending',  
  created_at TIMESTAMPTZ DEFAULT NOW(),  
  object_id  
);
```



lrs-schema-최종-wu.sql



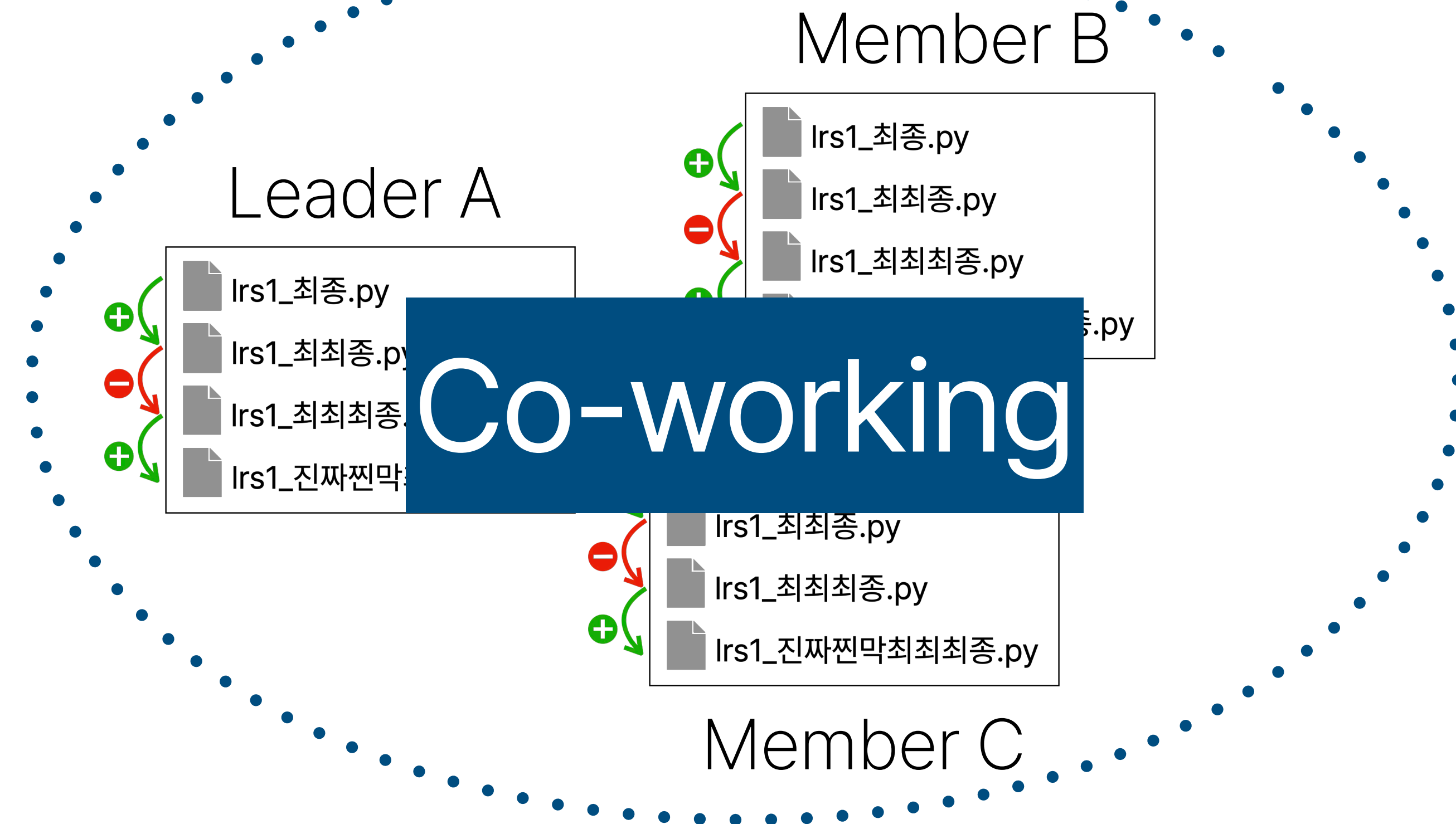
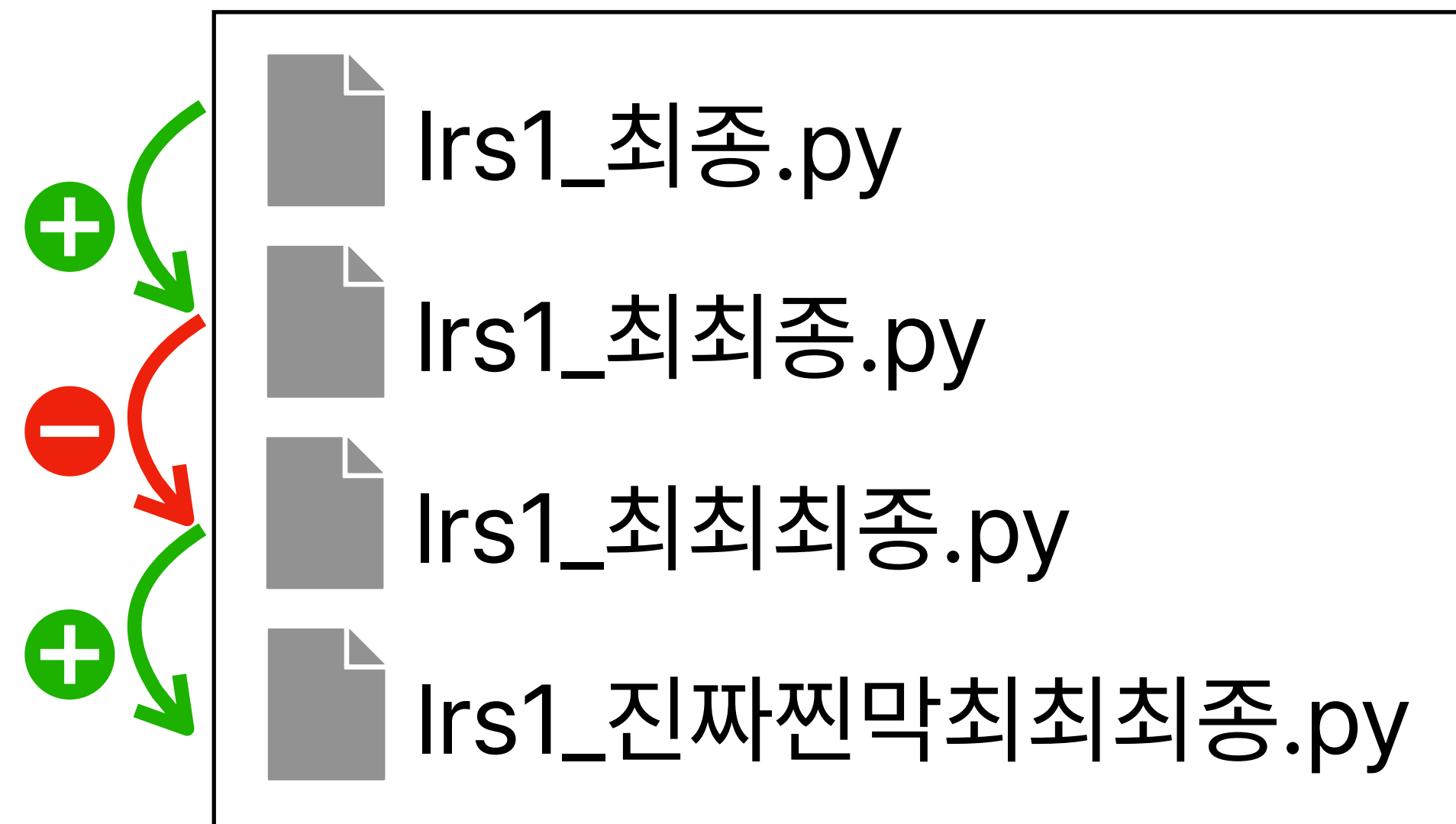
퇴사요 ~



Git & Github



Git & Github





Git & Github



Git & Github



→ 내 컴퓨터(local)에서의 버전 관리 → 내 컴퓨터(local)에서 설치 필요



Windows 11



[Install git for Windows 11](#)



메뉴얼 따라서 설치



명령 프롬프트(cmd)

```
git --version
```

*Reference : [git installation for win11](#)



MacOS



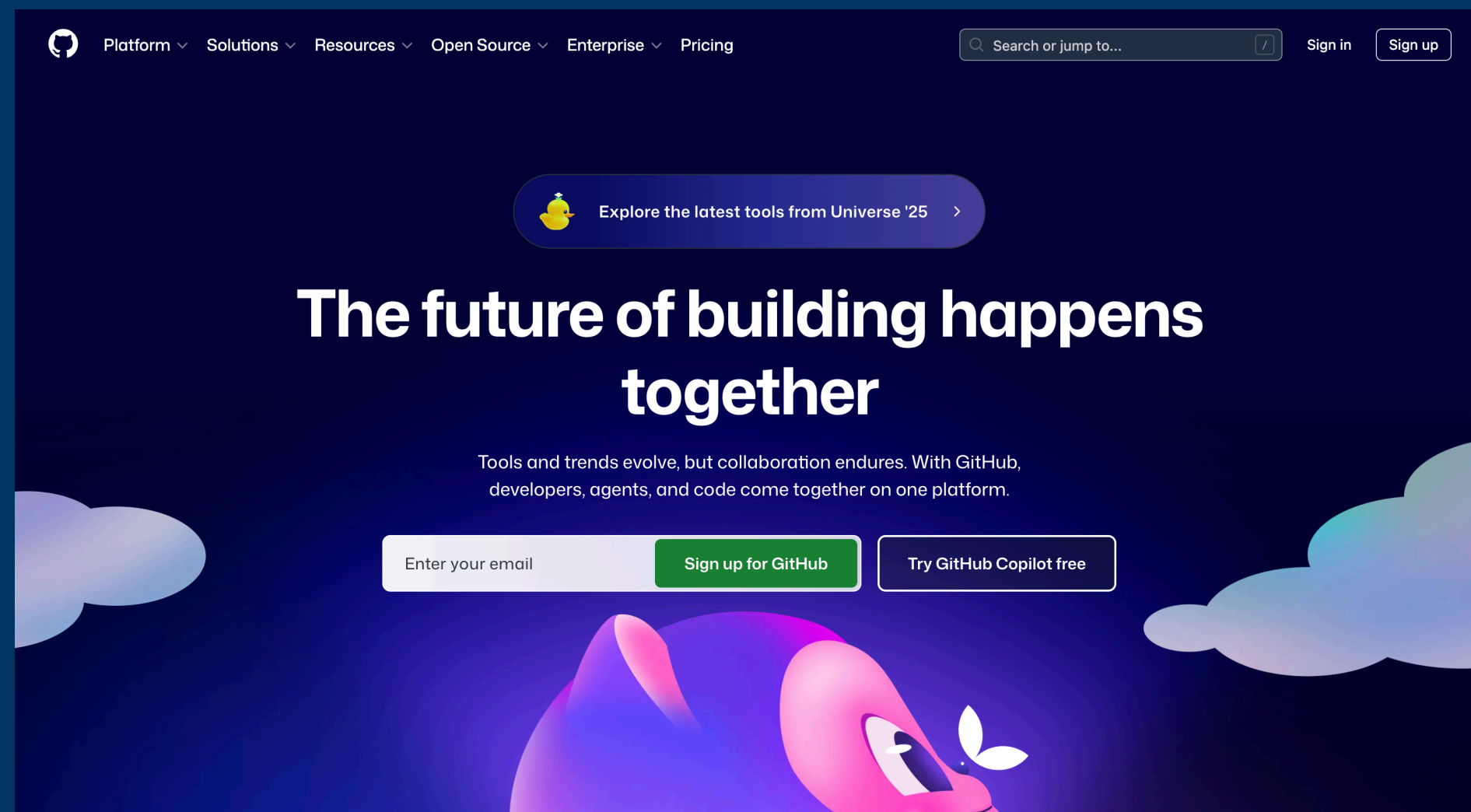
Terminal

```
git --version
```

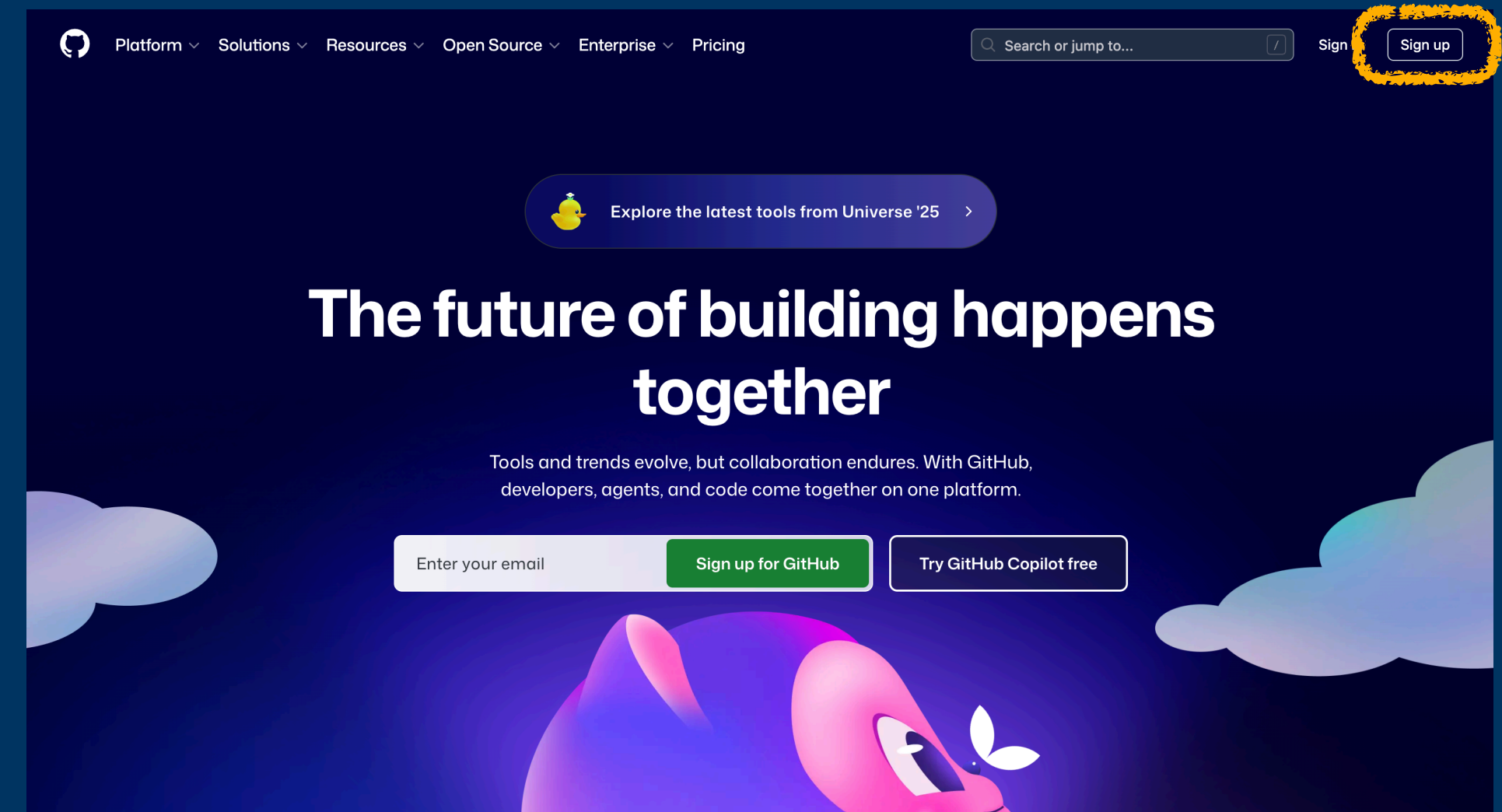



GitHub

→ 온라인 공간(Remote)에서의 협업 도구 → 웹사이트 가입



1. 🔗 Github 접속



2. 우측 상단 "Sign up"



GitHub

→ 온라인 공간(Remote)에서의 협업 도구 → 웹사이트 가입

Create your free account

Explore GitHub's core features for individuals and organizations.

See what's included ▾

Already have an account? [Sign in](#) →

Sign up for GitHub

Continue with Google

Continue with Apple

or

Email*

Email

Password*

Password

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

Username

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region*

Korea, South ▾

For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

Create account >

3. 원하는 방식으로 회원가입

*[Continue with Google](#) 권장

4. Dashboard 보이면 회원가입 완료

*상기 이미지는 예시 이미지로, 처음에는 아무것도 적혀있지 않음.

**Reference : [Github signup and create new repository](#)



Git & Github

Git & Github

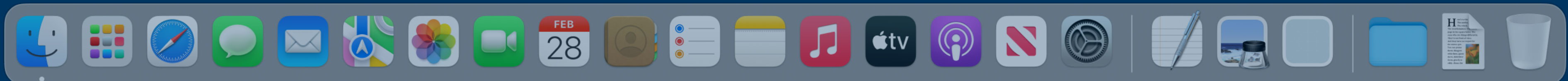


Terminal



2026-lrs

```
Last login: Wed Dec 31 02:53:44 on ttys007
RainyForest@ ~ ▶ git config --global user.name "깃허브아이디"
RainyForest@ ~ ▶ git config --global user.email "깃허브이메일"
RainyForest@ ~ ▶ cd Desktop // 프로젝트 폴더를 만들고자 하는 위치로 이동(change dir)
RainyForest@ ~/Desktop ▶ mkdir 2026-lrs // 프로젝트 폴더 생성(make dir)
RainyForest@ ~/Desktop ▶ cd 2026-lrs // 만든 폴더로 이동(change dir)
RainyForest@ ~/Desktop/2026-lrs ▶ git init // git 저장소 만들기 위한 초기화
```



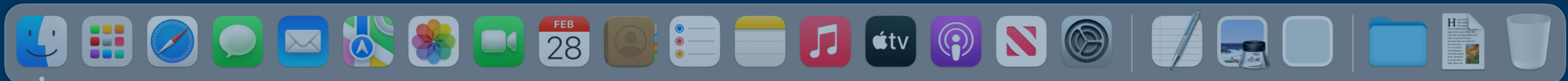
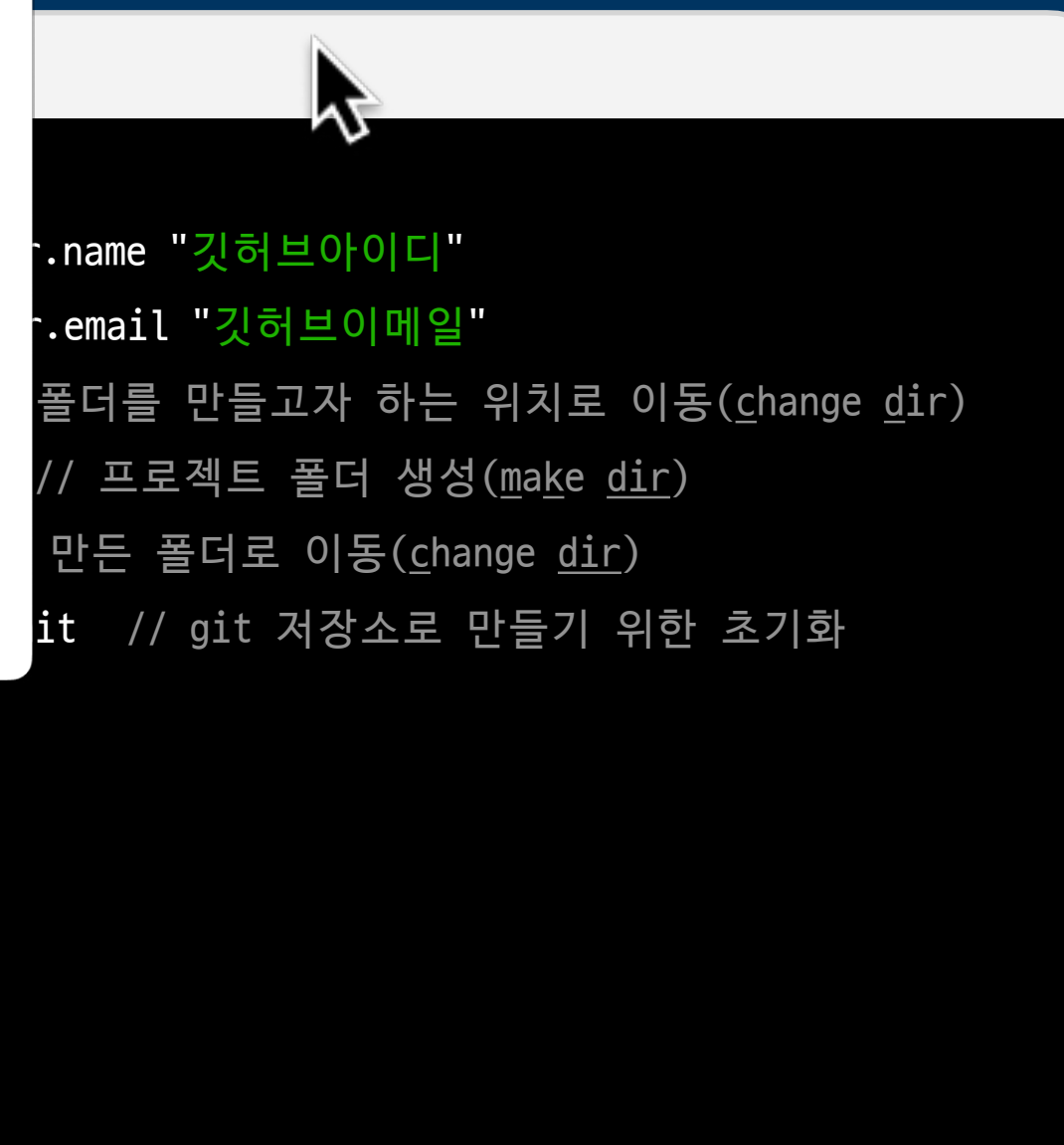
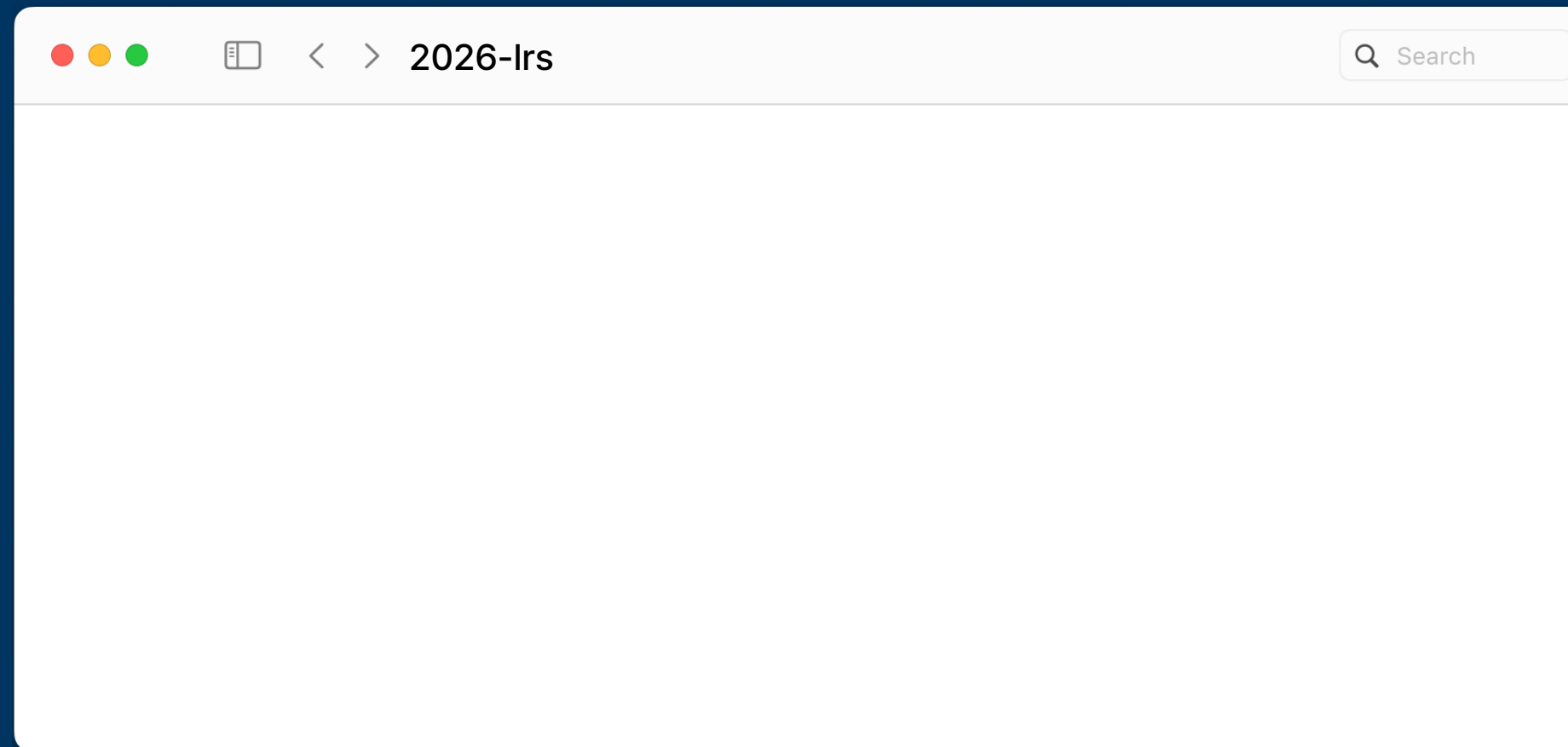
Git & Github



Terminal



2026-lrs



Git & Github



Terminal



2026-lrs

```
Last login: Wed Dec 31 02:53:44 on ttys007
RainyForest@ ~ % git config --global user.name RainyForest
RainyForest@ ~ % git config --global user.email rainyforest23@gmail.com
RainyForest@ ~ % cd Desktop // 프로젝트 폴더로 이동
RainyForest@ ~/Desktop % mkdir 2026-lrs // 2026-lrs 폴더 생성
RainyForest@ ~/Desktop % cd 2026-lrs // 2026-lrs 폴더로 이동
RainyForest@ ~/Desktop/2026-lrs % git init
```

RainyForest23

Q Type to search

+ -

Overview

Repositories 24

Projects

Packages

Stars

Woorim Shin

RainyForest23

Undergraduate student interested in AI/ML.

Edit profile

1 follower · 3 following

RainyForest23 / README.md

About Me

I'm currently pursuing my degree in **Computer Science and Engineering** at **Ewha Womans University**. With a solid foundation in computer science fundamentals, I'm particularly passionate about **AI/ML** and exploring how technology can solve real-world problems.

I've had the opportunity to lead and contribute to several high school education projects at **Hiconsy**, an EdTech startup in South Korea, where I've gained valuable experience in project management and educational technology development.

Recently, I led the end-to-end development of an **AI subtitle generation model for Korean educational content**. Through this project, I gained comprehensive experience in:

- Model Development:** Fine-tuning Whisper models with LoRA and PEFT techniques
- Infrastructure:** Setting up distributed GPU training environments and monitoring with Tensorboard



Git & Github



Terminal



2026-lrs

```
Last login: Wed Dec 31 02:53:44 on ttys007
RainyForest@ ~ % git config --global user.
RainyForest@ ~ % git config --global user.
RainyForest@ ~ % cd Desktop // 프로젝트
RainyForest@ ~/Desktop % mkdir 2026-lrs /
RainyForest@ ~/Desktop % cd 2026-lrs //
RainyForest@ ~/Desktop/2026-lrs % git ini
```

New repository

Q Type / to search

+ 🔍 📄 📁 🔔

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

RainyForest23

Repository name *

2026-lrs

Great repository names are short and memorable. How about [musical-fishstick](#)?

Description

0 / 350 characters

2

Configuration

Choose visibility *

Public

Choose who can see and commit to this repository

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

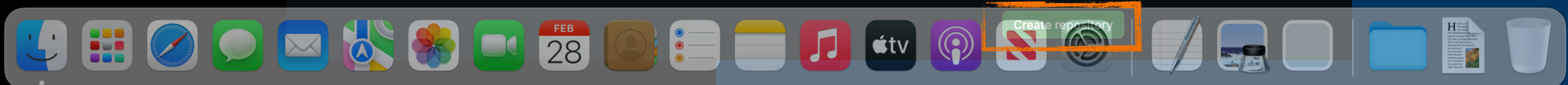
.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

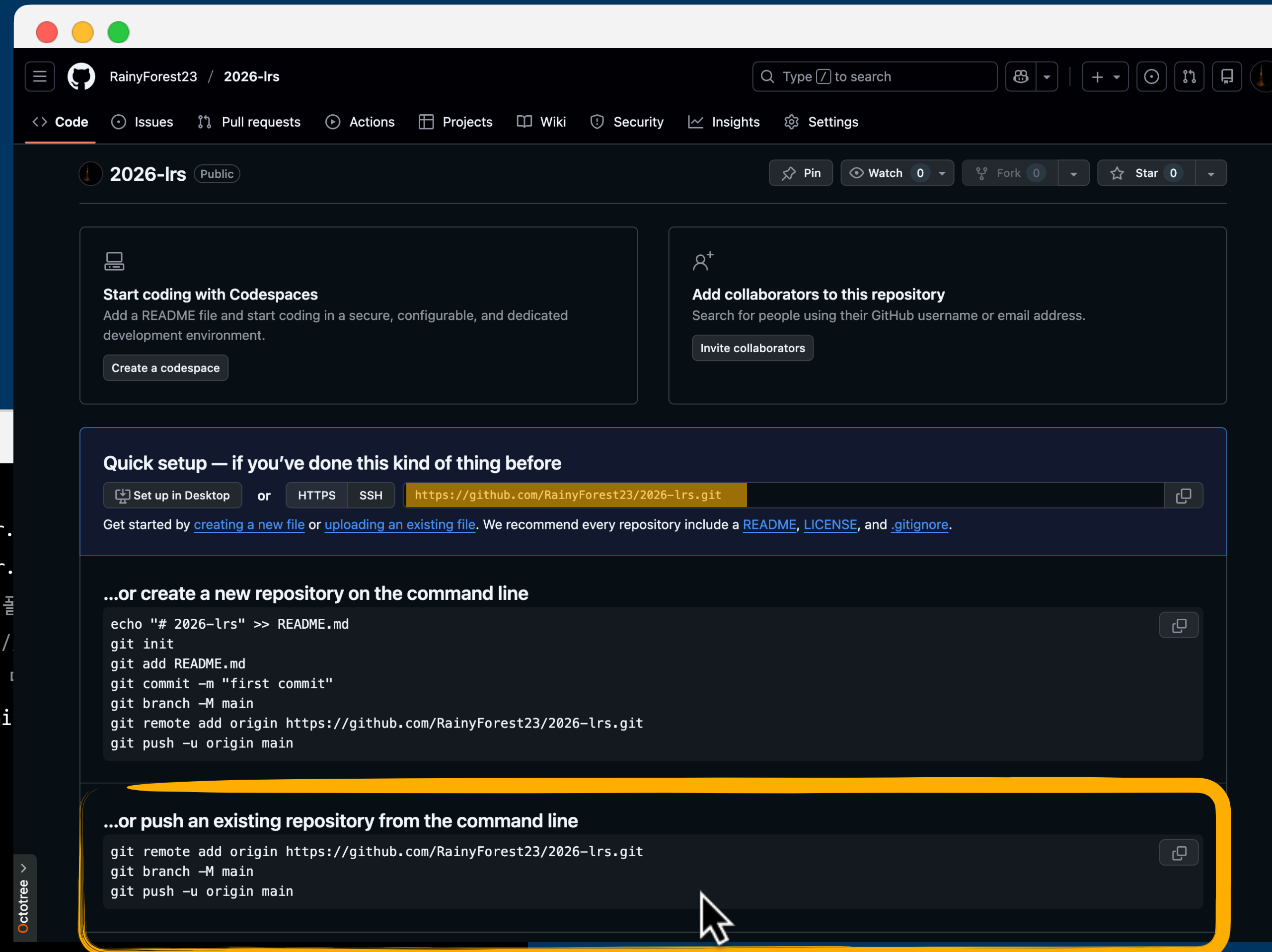
No license





2026-lrs

```
Last login: Wed Dec 31 02:53:44 on ttys007
RainyForest@ ~ % git config --global user.
RainyForest@ ~ % git config --global user.
RainyForest@ ~ % cd Desktop // 프로젝트폴
RainyForest@ ~/Desktop % mkdir 2026-lrs /
RainyForest@ ~/Desktop % cd 2026-lrs //
RainyForest@ ~/Desktop/2026-lrs % git ini
```



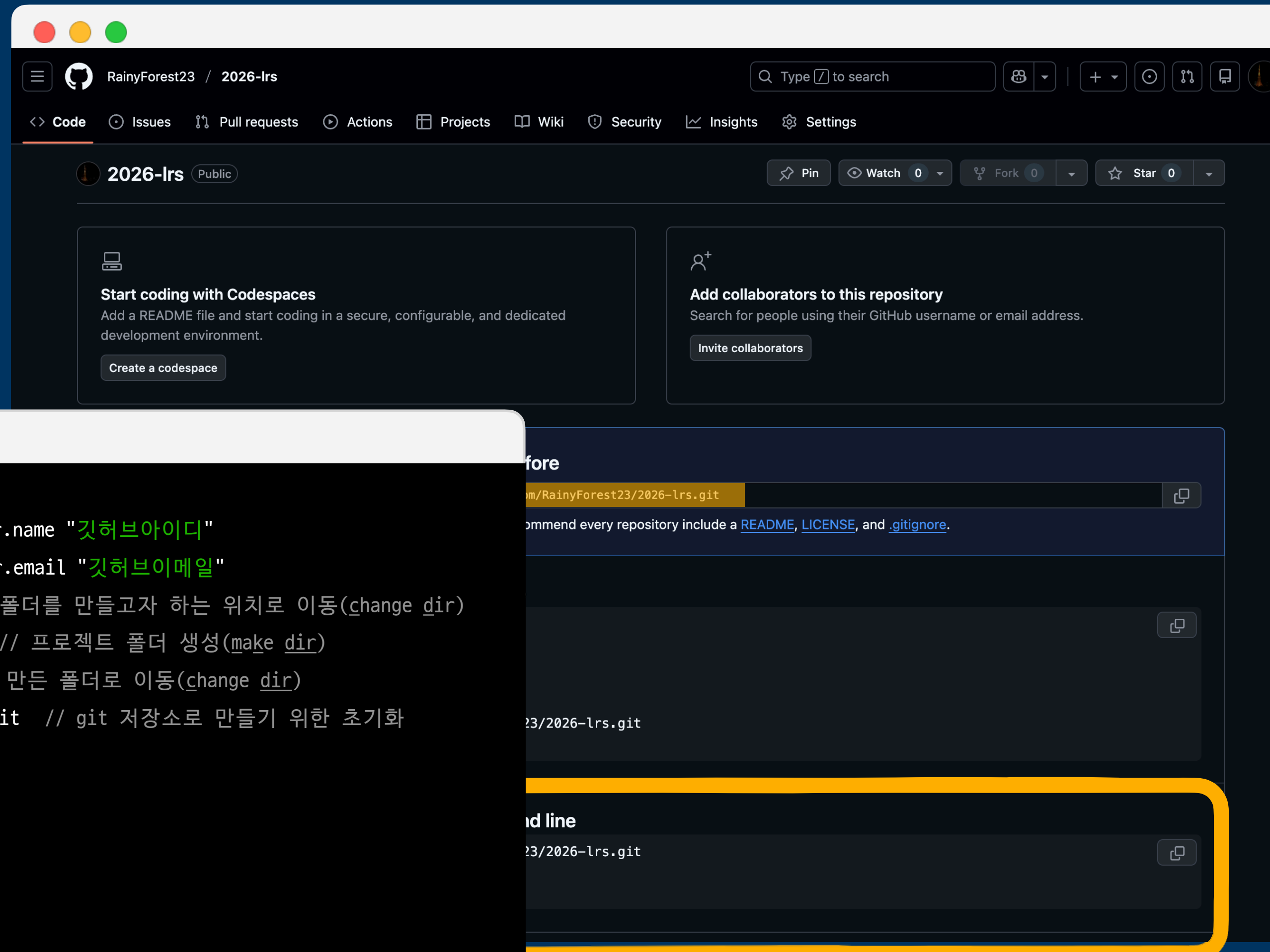
Git & Github



Terminal



2026-lrs



```
Last login: Wed Dec 31 02:53:44 on ttys007
RainyForest@ ~ % git config --global user.name "깃허브아이디"
RainyForest@ ~ % git config --global user.email "깃허브이메일"
RainyForest@ ~ % cd Desktop // 프로젝트 폴더를 만들고자 하는 위치로 이동(change dir)
RainyForest@ ~/Desktop % mkdir 2026-lrs // 프로젝트 폴더 생성(make dir)
RainyForest@ ~/Desktop % cd 2026-lrs // 만든 폴더로 이동(change dir)
RainyForest@ ~/Desktop/2026-lrs % git init // git 저장소로 만들기 위한 초기화
```

Git & Github



Terminal



2026-lrs

```

RainyForest23 / 2026-lrs
Type to search
Code Issues Pull requests Actions Projects Wiki Security Insights Settings
2026-lrs Public Pin Watch 0 Fork 0 Star 0
Last login: Wed Dec 31 02:53:44 on ttys007
RainyForest@ ~ % git config --global user.name "깃허브아이디"
RainyForest@ ~ % git config --global user.email "깃허브이메일"
RainyForest@ ~ % cd Desktop // 프로젝트 폴더를 만들고자 하는 위치로 이동(change dir)
RainyForest@ ~/Desktop % mkdir 2026-lrs // 프로젝트 폴더 생성(make dir)
RainyForest@ ~/Desktop % cd 2026-lrs // 만든 폴더로 이동(change dir)
RainyForest@ ~/Desktop/2026-lrs % git init // git 저장소로 만들기 위한 초기화
RainyForest@ ~/Desktop/2026-lrs % git remote add origin "복사한 링크"
RainyForest@ ~/Desktop/2026-lrs % git branch -M main
RainyForest@ ~/Desktop/2026-lrs % git push -u origin main

...or push an existing repository from the command line
git remote add origin https://github.com/RainyForest23/2026-lrs.git
git branch -M main
git push -u origin main
```


Git & Github

Create repository

2026-lrs

Internet / Cloud

Last login: Wed Dec 31 02:53:44 on ttys007

RainyForest@ ~ `git config --global user.name "깃허브 아이디"`

RainyForest@ ~ `git config --global user.email "깃허브 이메일"`

RainyForest@ ~ `cd Desktop` // 프로젝트 폴더를 만들고자 하는 위치로 이동(change dir)

RainyForest@ ~/Desktop `mkdir 2026-lrs` // 프로젝트 폴더 생성(make dir)

RainyForest@ ~/Desktop `cd 2026-lrs` // 만든 폴더로 이동(change dir)

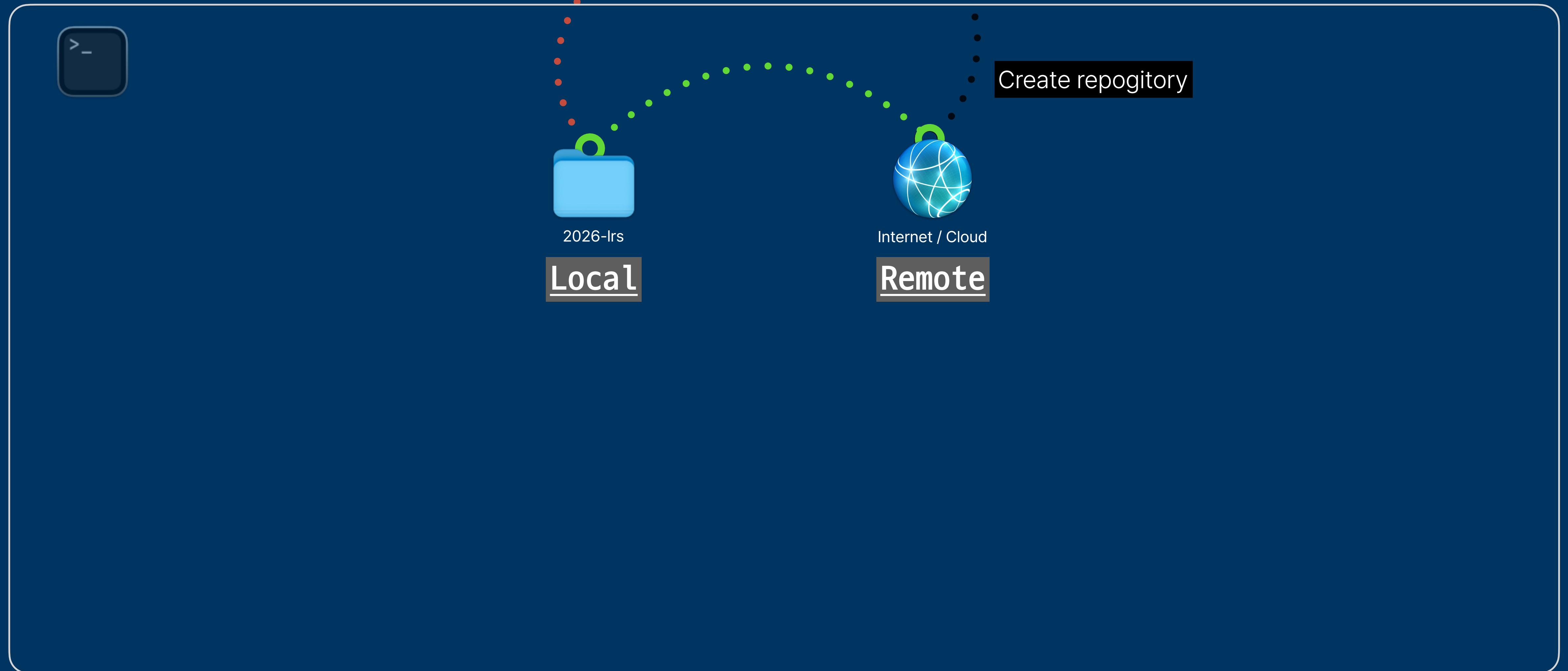
RainyForest@ ~/Desktop/2026-lrs `git init` // git 저장소로 만들기 위한 초기화

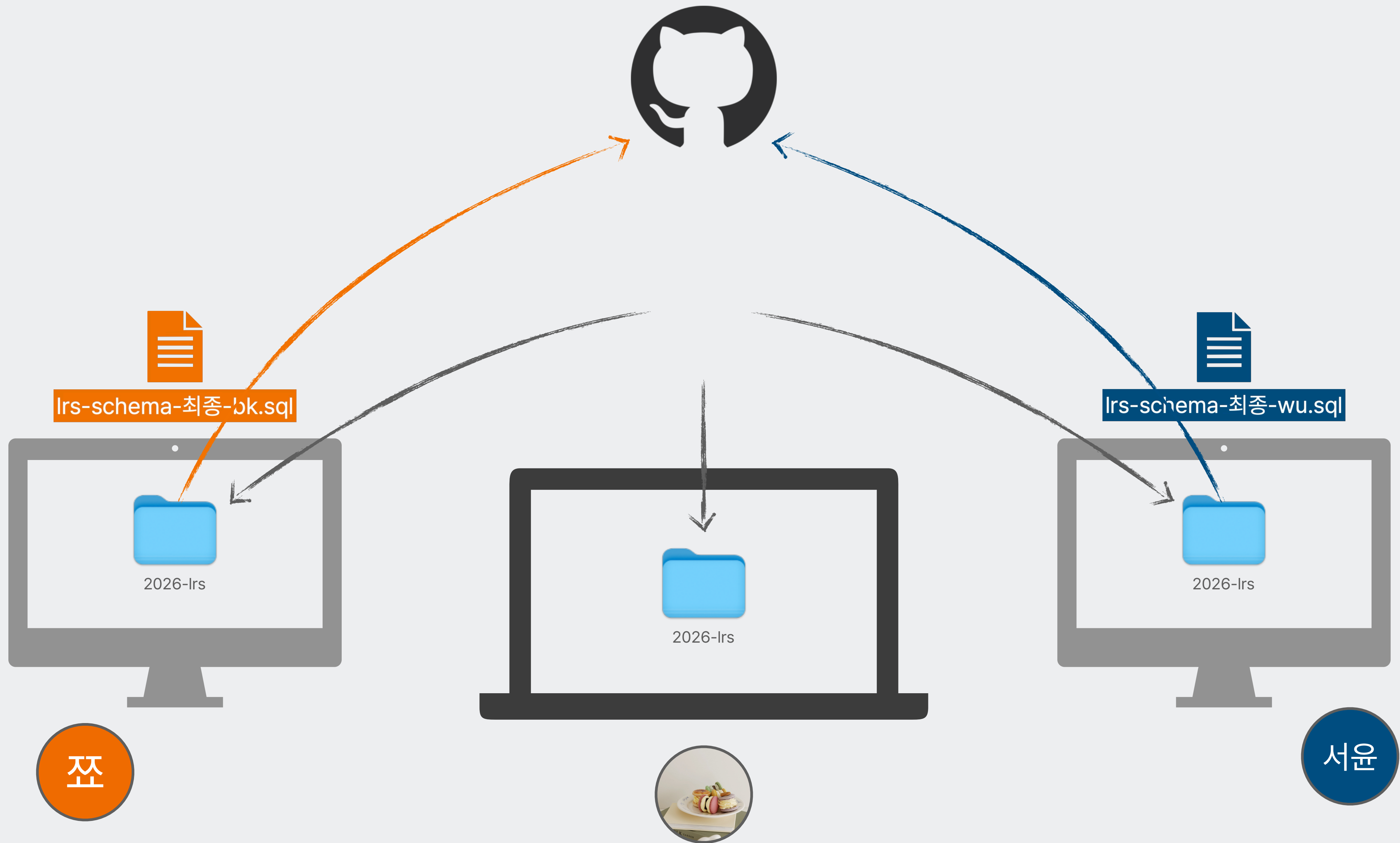
RainyForest@ ~/Desktop/2026-lrs `git remote add origin "복사한 링크"`

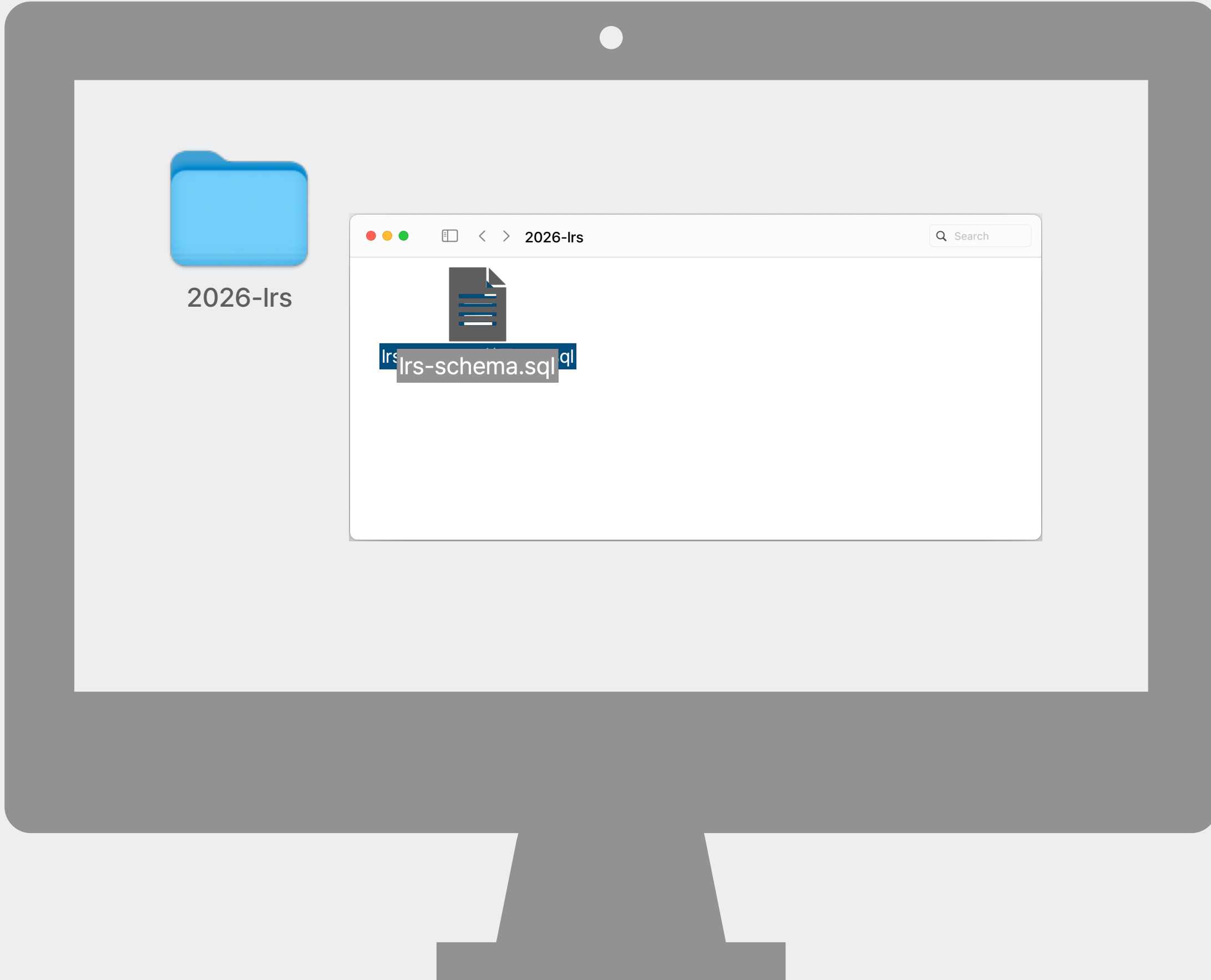
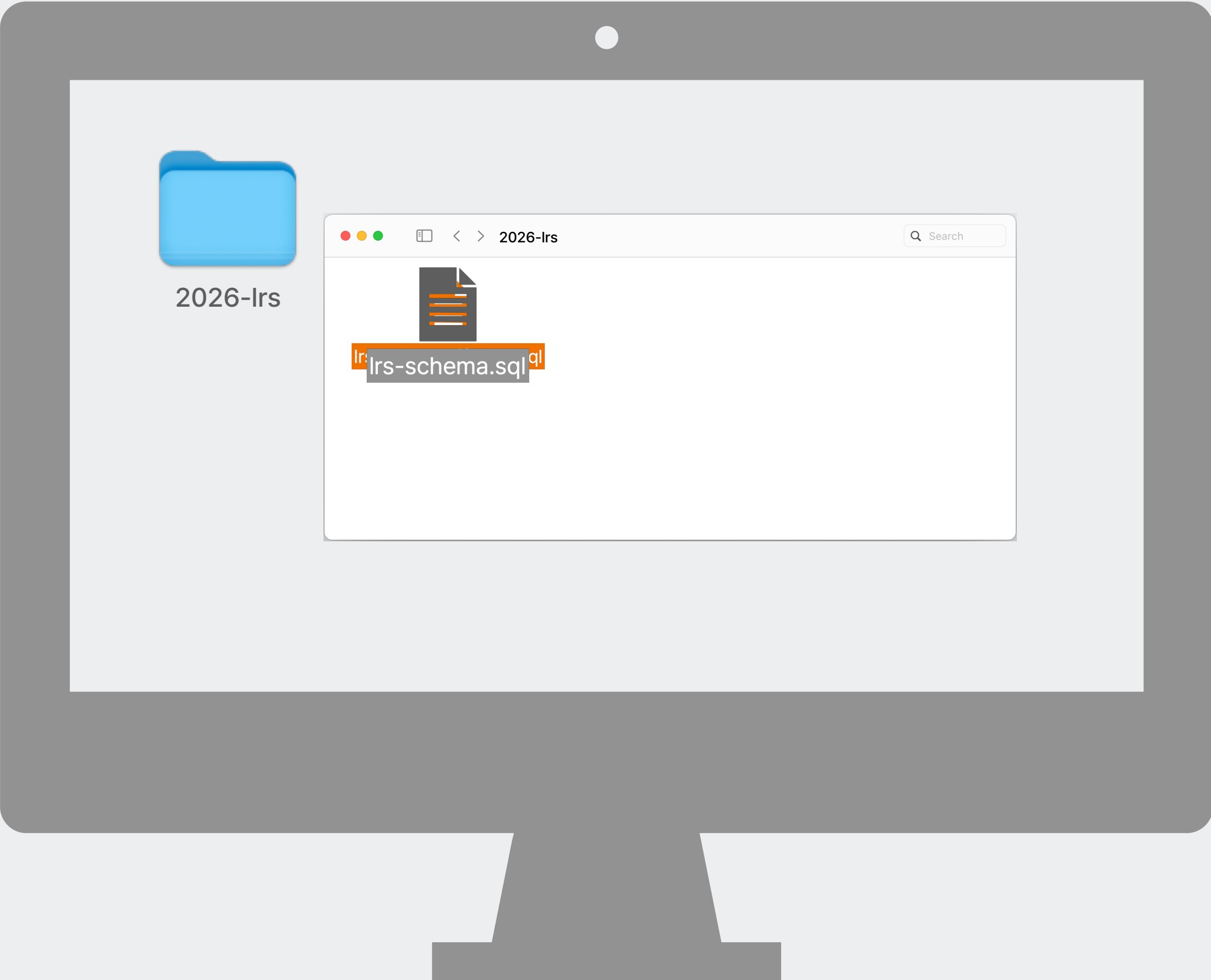
RainyForest@ ~/Desktop/2026-lrs `git branch -M main`

RainyForest@ ~/Desktop/2026-lrs `git push -u origin main`

Git & Github







feat : 기능 추가
fix : 버그 수정
refactor : 리팩토링
docs : 문서 수정
test : 테스트 또는 테스트 코드 추가

\$ git push origin main

\$ git push origin main

\$ git commit -m "커밋메시지"

\$ git commit -m "커밋메시지"

lrs-schema-최종-bk.sql

\$ git add .

lrs-schema-최종-wu.sql

\$ git add .

2026-lrs

2026-lrs

2026-lrs

쫐

서윤



뭔가 변경한게 많은 것 같을 때..
remote repo 변경 사항 가져오기만

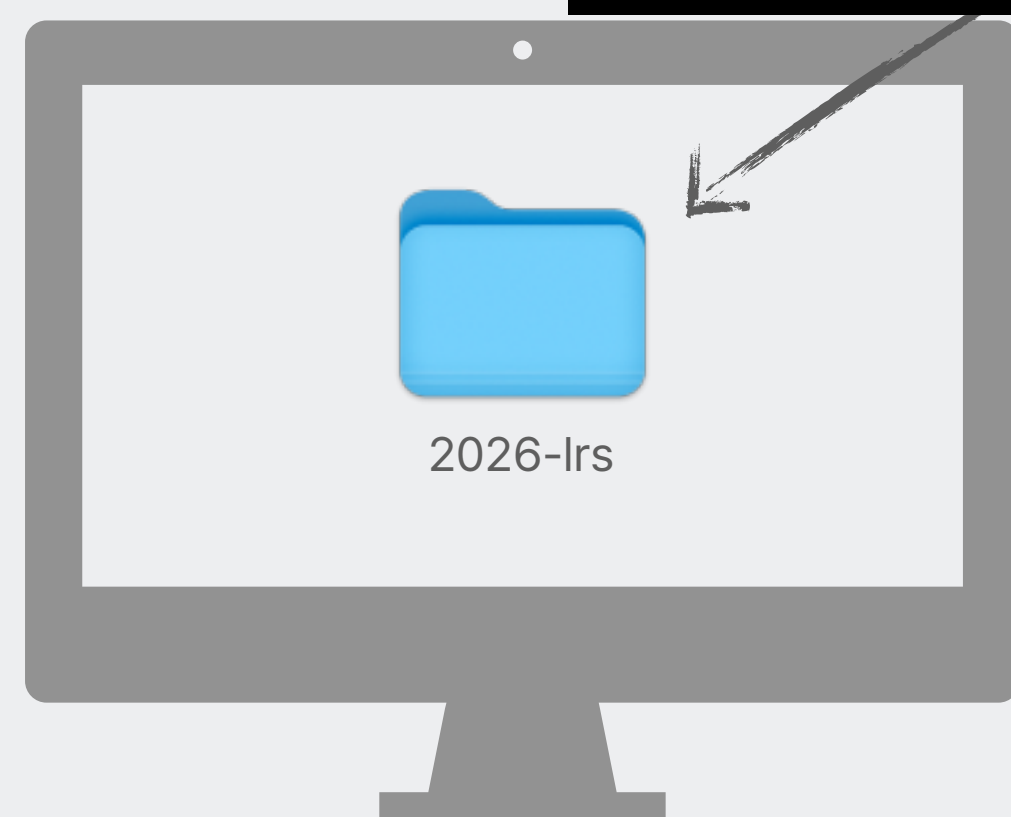
pull = fetch + merge

remote repo 변경 사항 가져오고 합치기
딱히 변경한거 많이 없을 것 같을 때..

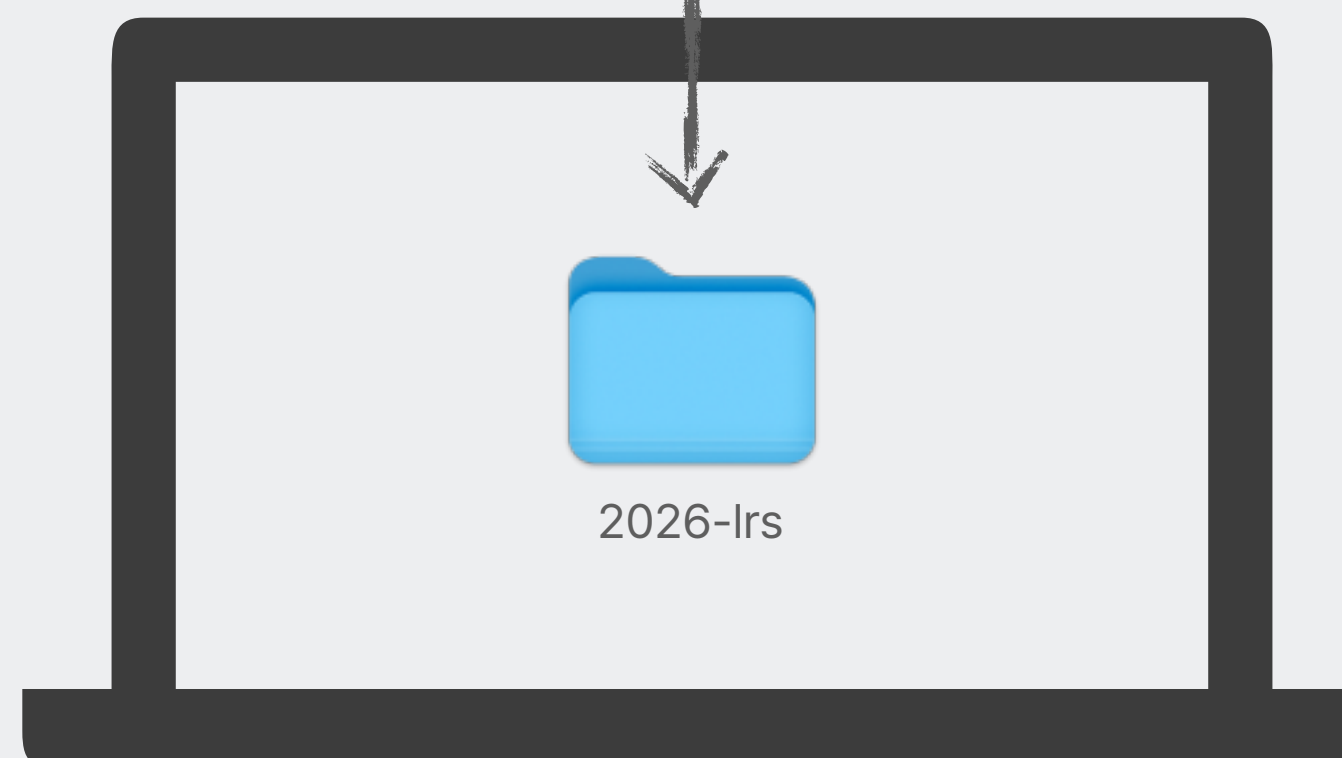


lrs-schema.sql

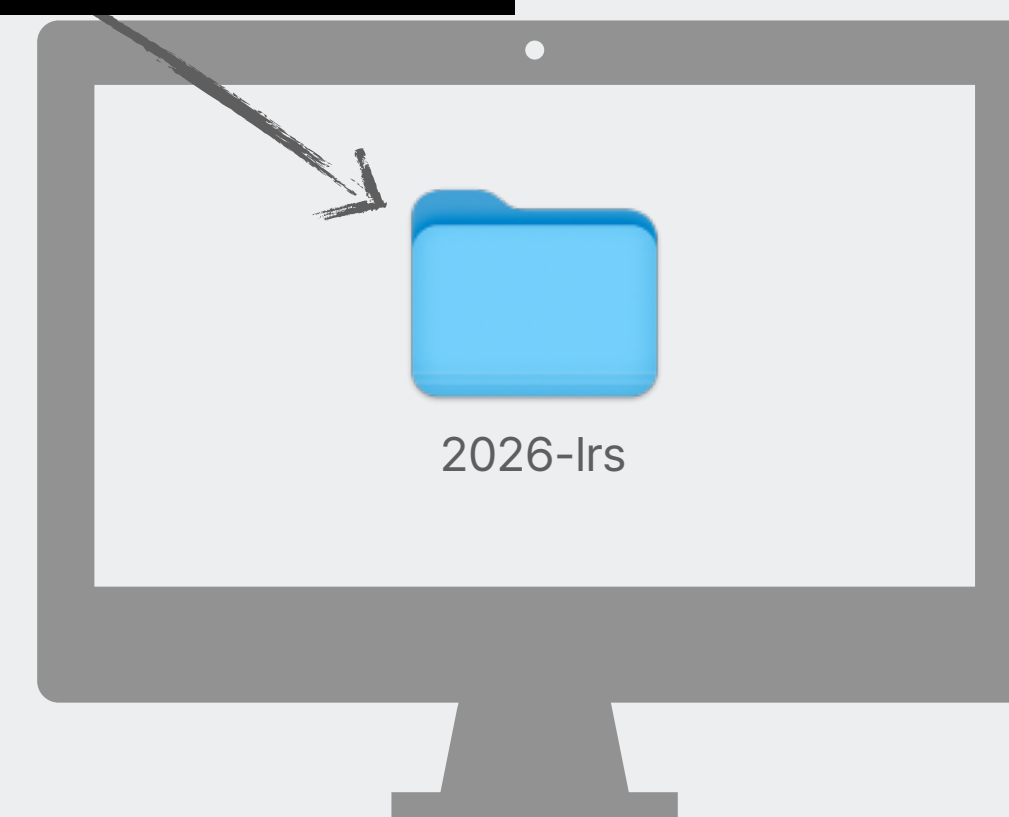
```
$ git pull origin main  
$ git fetch origin main
```



```
$ git pull origin main  
$ git fetch origin main
```



```
$ git pull origin main  
$ git fetch origin main
```



RainyForest23 / 2025-LRS

Type to search

<> Code

Pull requests

Actions

Projects

Security

Insights

Settings

🔔 2025-LRS

Public

forked from SDIJ-Forest-Lecture/2025-LRS

🔑 main

🔑 1 Branch

🏷️ 0 Tags

🔍 Go to file

⌕

Add file

<> Code

This branch is up to date with SDIJ-Forest-Lecture/2025-LRS:main

🔗 Contribute

↻ Sync fork

🔔 RainyForest23

Fix: RLS issue in signup and redirect location

9f3ea94 · 2 weeks ago

🕒 28 Commits

📁 public	feat: 프로젝트 초기 설정 및 필수 패키지 설치 (Phase 1.1-1.2)	2 weeks ago
📁 scripts	Feat: Implement Admin User Management and Initial Seed	2 weeks ago
📁 src	Fix: RLS issue in signup and redirect location	2 weeks ago
📁 supabase	Fix admin role recognition and RLS infinite recursion issues	2 weeks ago
📄 .env.example	feat: 프로젝트 초기 설정 및 필수 패키지 설치 (Phase 1.1-1.2)	2 weeks ago
📄 .gitignore	feat: Supabase 연결 설정 완료 (Phase 1.4)	2 weeks ago
📄 README.md	feat: 프로젝트 초기 설정 및 필수 패키지 설치 (Phase 1.1-1.2)	2 weeks ago
📄 SETUP.md	feat: 프로젝트 초기 설정 및 필수 패키지 설치 (Phase 1.1-1.2)	2 weeks ago
📄 components.json	feat: 프로젝트 초기 설정 및 필수 패키지 설치 (Phase 1.1-1.2)	2 weeks ago
📄 debug-user-role.js	Fix admin role recognition and RLS infinite recursion issues	2 weeks ago
📄 eslint.config.mjs	feat: 프로젝트 초기 설정 및 필수 패키지 설치 (Phase 1.1-1.2)	2 weeks ago

About

ⓘ LRS-Dev

📖 Readme

📈 Activity

⭐ 0 stars

👁️ 0 watching

🍴 0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

TypeScript 87.9%

PLpgSQL 7.6%

CSS 2.3%

JavaScript 2.2%

Suggested workflows

RainyForest23 / 2025-LRS

Type to search

<> Code

Pull requests

Actions

Projects

Security

Insights

Settings

Commits

🔑 main

🔍 All users

📅 All time

📅 Commits on Dec 27, 2025

Fix: RLS issue in signup and redirect location

🔔 RainyForest23 committed 2 weeks ago

9f3ea94

📄 <>

Fix: Allow signup route in middleware

🔔 RainyForest23 committed 2 weeks ago

e61c1c6

📄 <>

Fix admin role recognition and RLS infinite recursion issues

🔔 RainyForest23 and claude committed 2 weeks ago

64ecee0

📄 <>

Fix: Syntex error in layout and login form

🔔 RainyForest23 committed 2 weeks ago

a841486

📄 <>

Feat: Add signup link to login form

🔔 RainyForest23 committed 2 weeks ago

6a58e88

📄 <>

Feat: Implement Sign Up Page and Logic

🔔 RainyForest23 committed 2 weeks ago

023afd3

📄 <>

Debug: Add role indicator to NavMenu

🔔 RainyForest23 committed 2 weeks ago

dc802ca

📄 <>

Refactor: Update Admin Layout with dark sidebar

🔔 RainyForest23 committed 2 weeks ago

b48040e

📄 <>

Add navigation menus to dashboard and admin layouts

🔔 RainyForest23 and claude committed 2 weeks ago

fdf7cb8

📄 <>

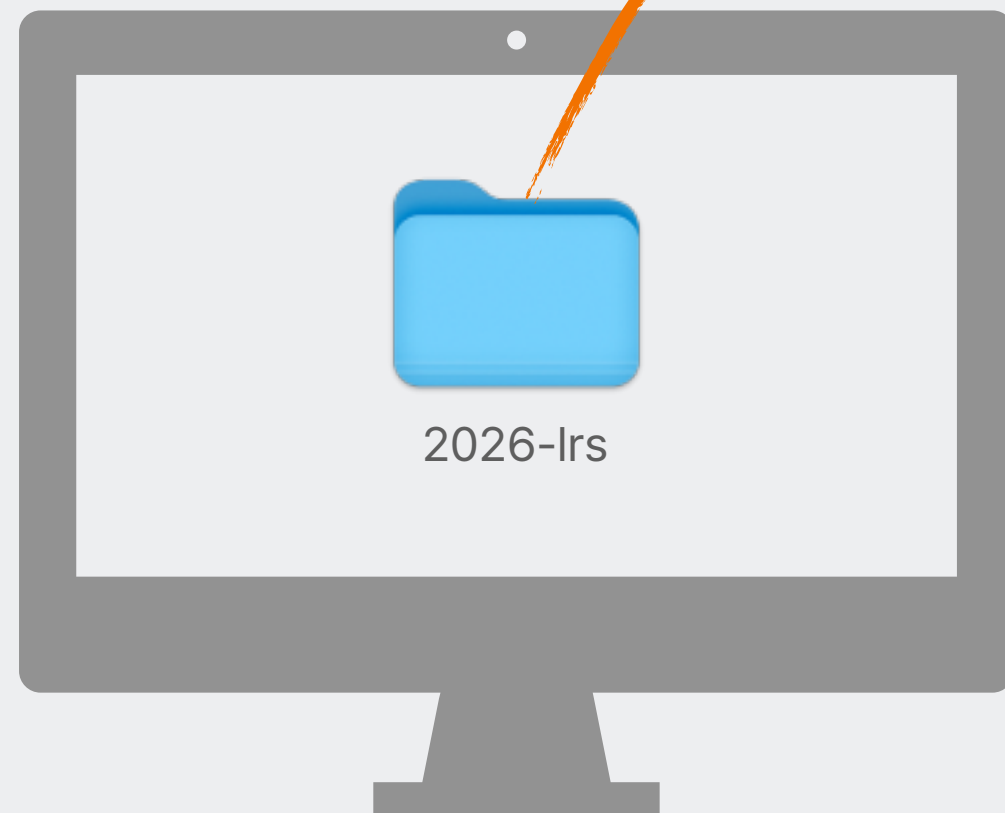


좀 골치아파짐...

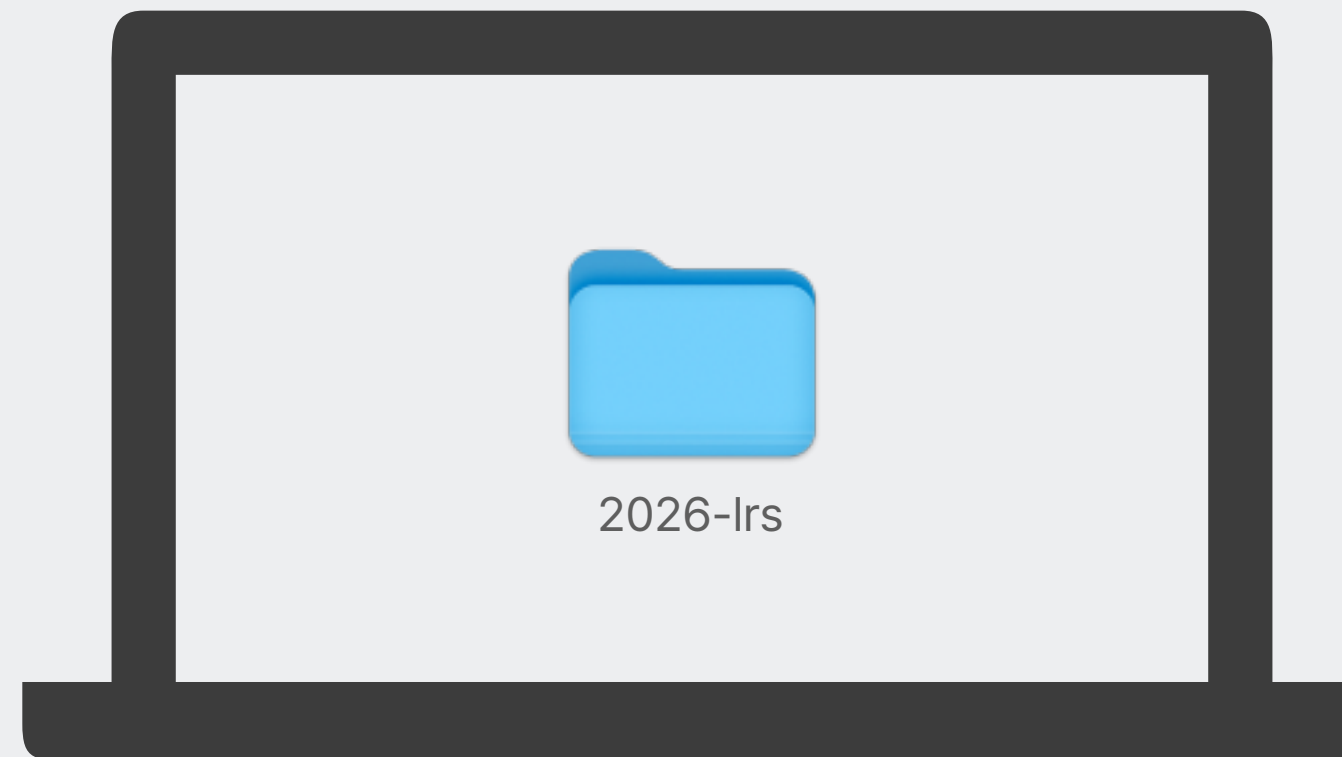
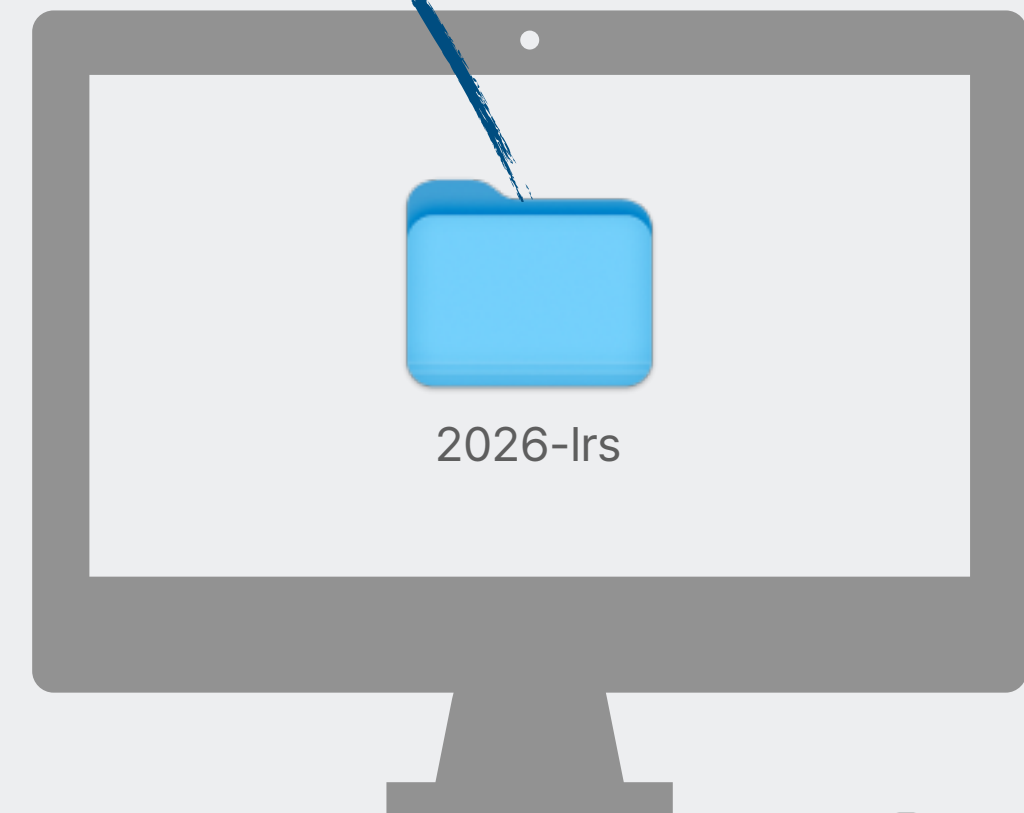
!! commit conflict !!



lrs-schema-최종-bk.sql

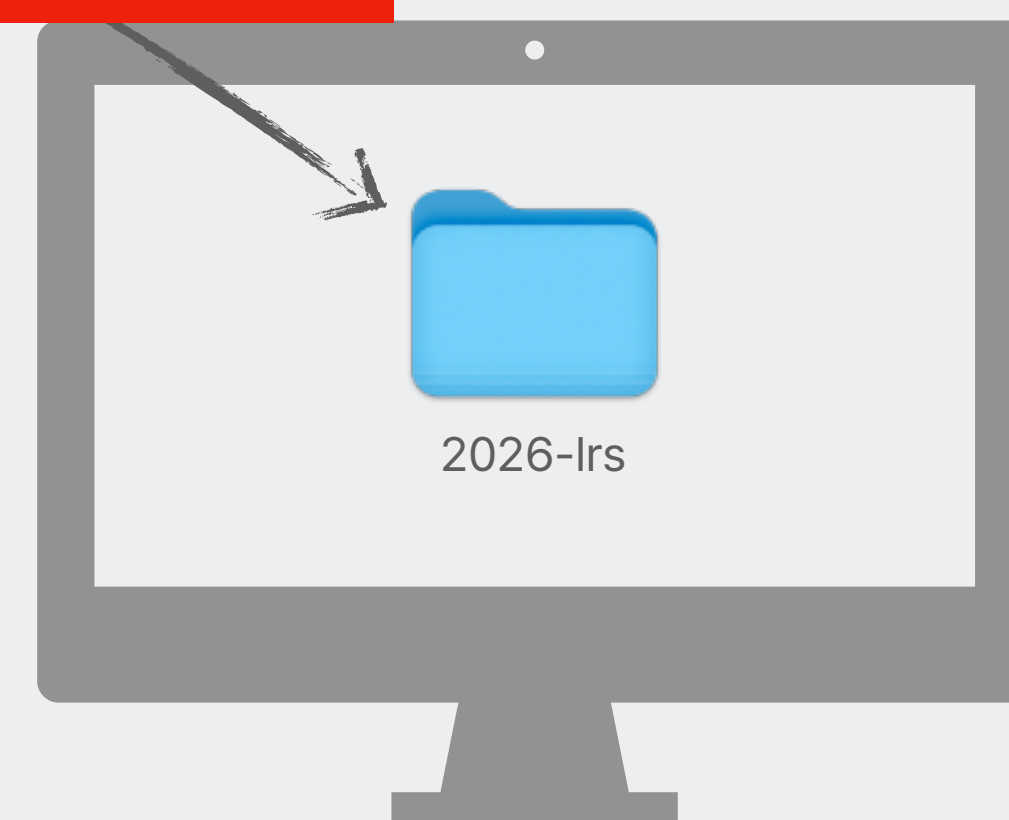
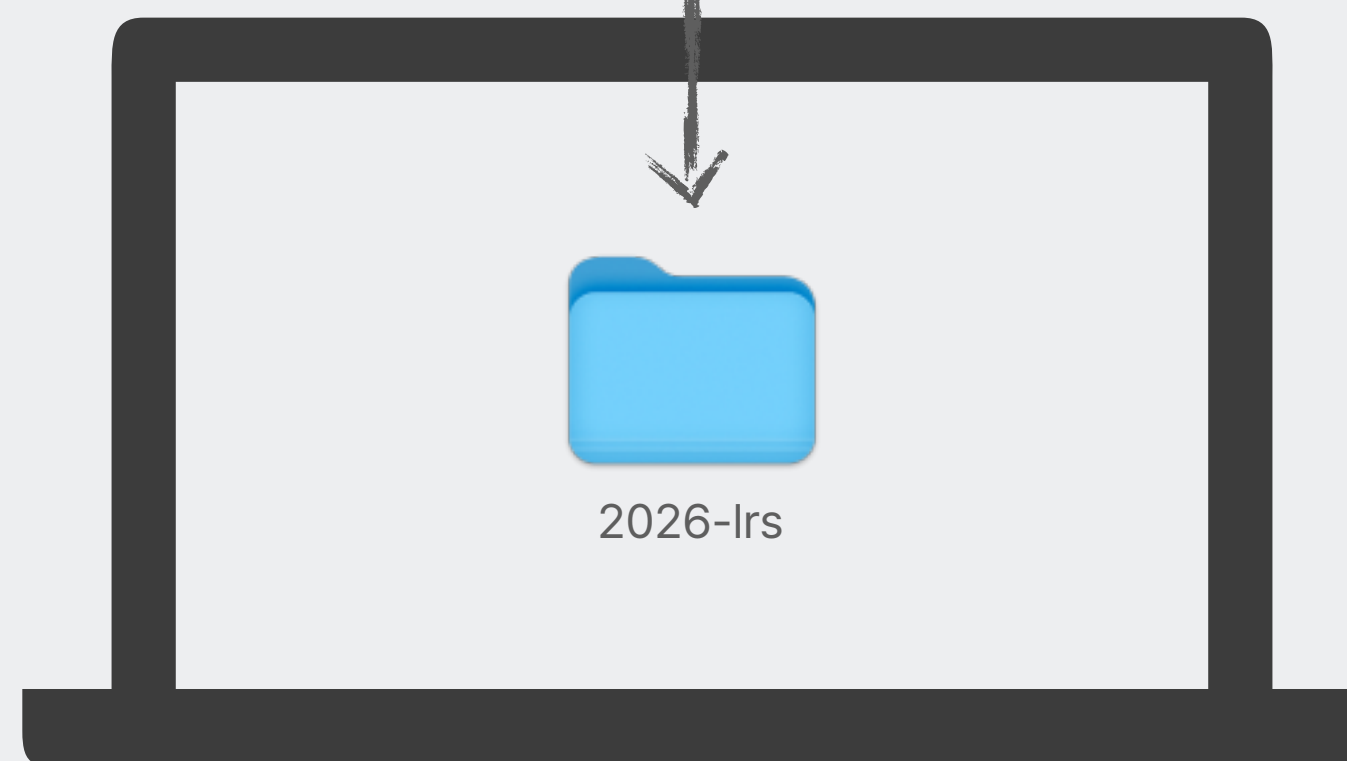
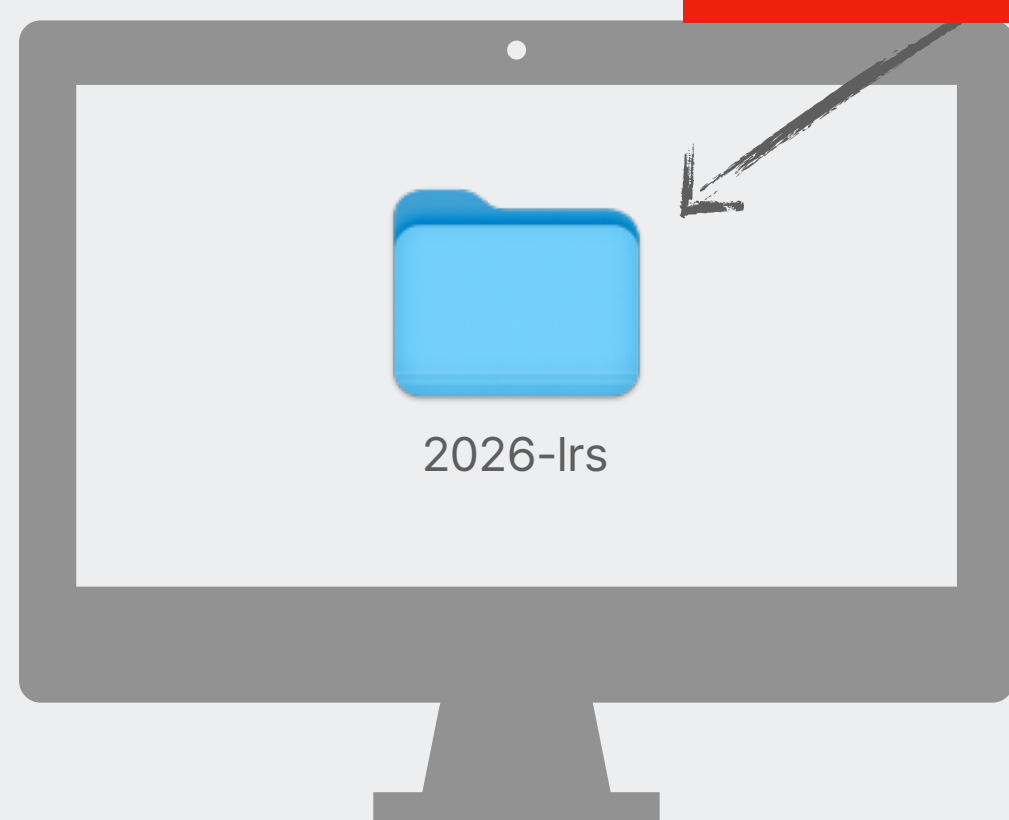


lrs-schema-최종-wu.sql



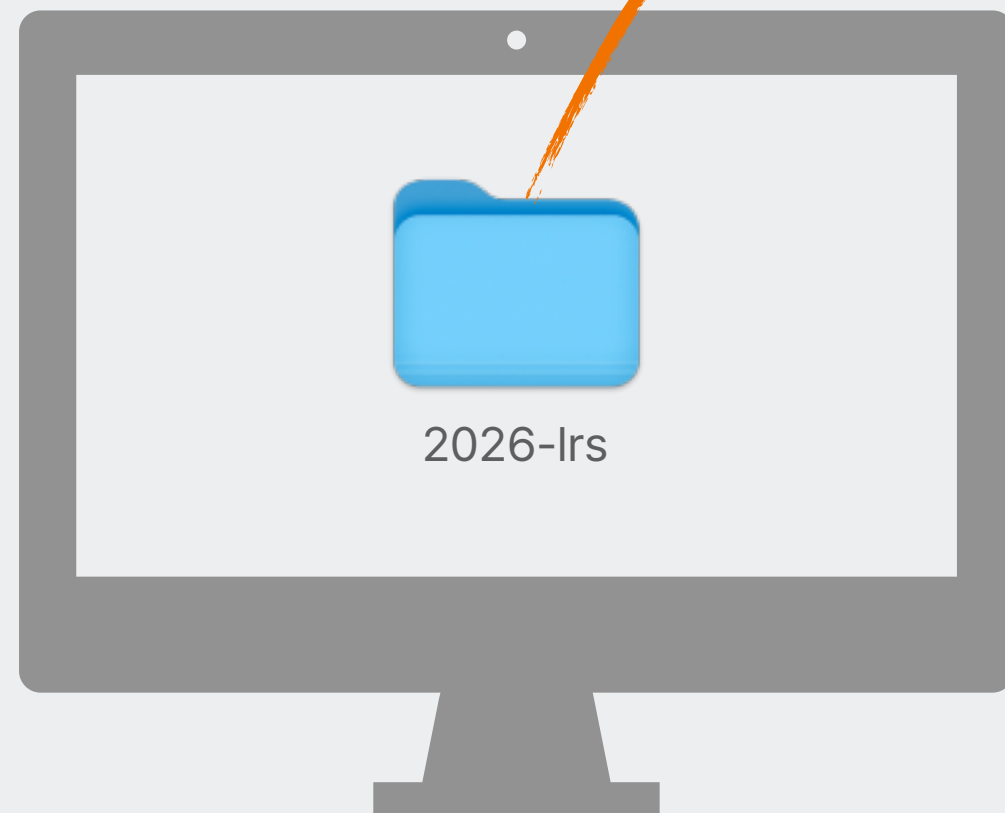


commit 하기 전..
fetch or pull 했나요?

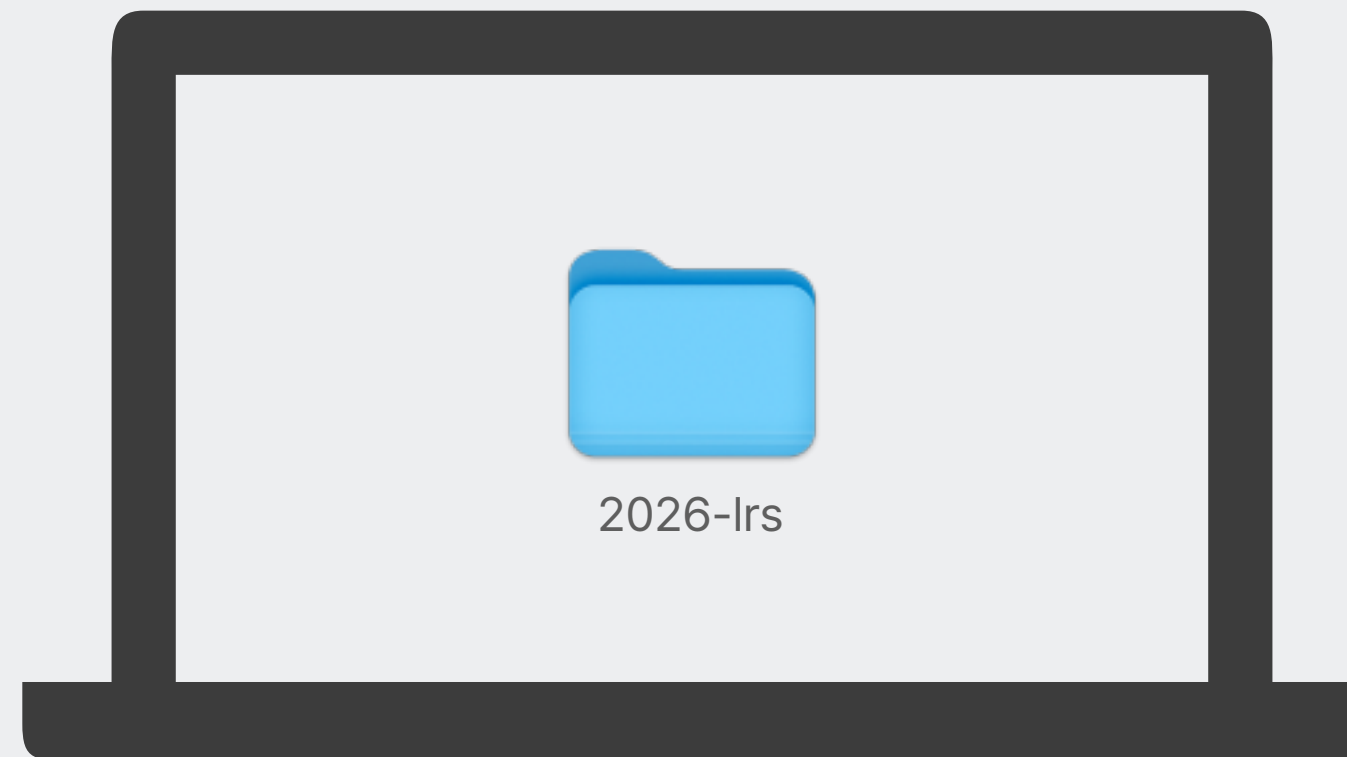
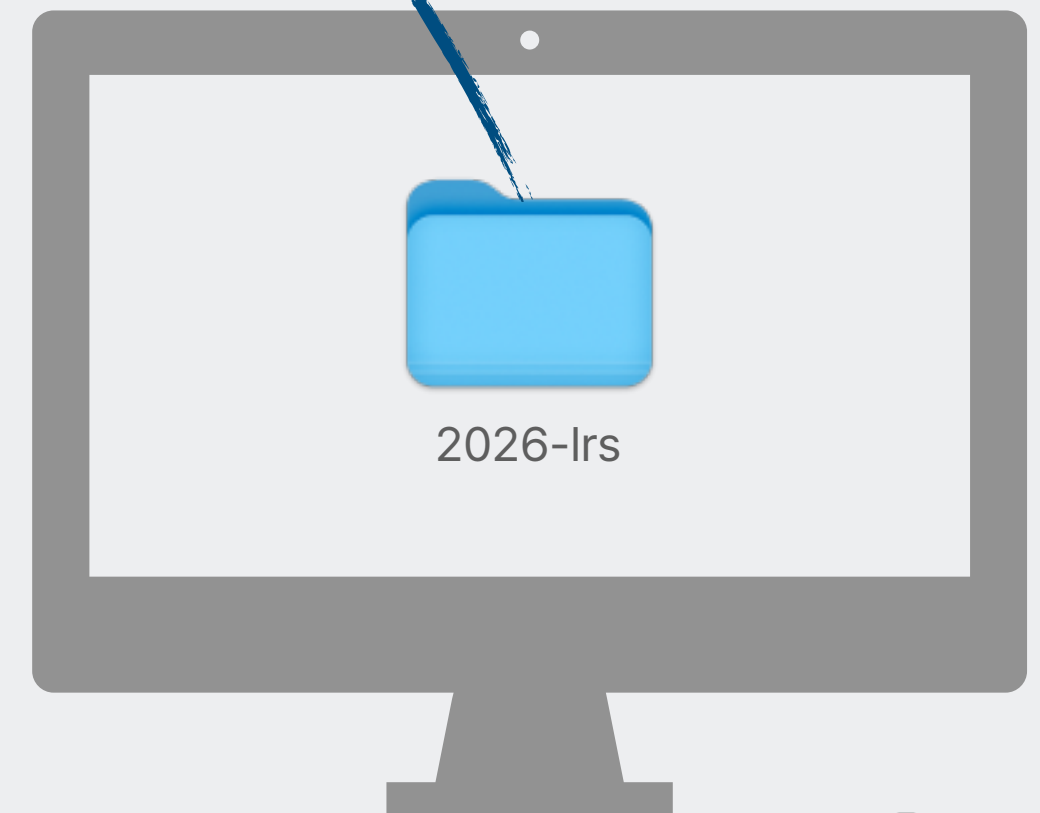




lrs-schema-최종-bk.sql



lrs-schema-최종-wu.sql





***branch**

merge

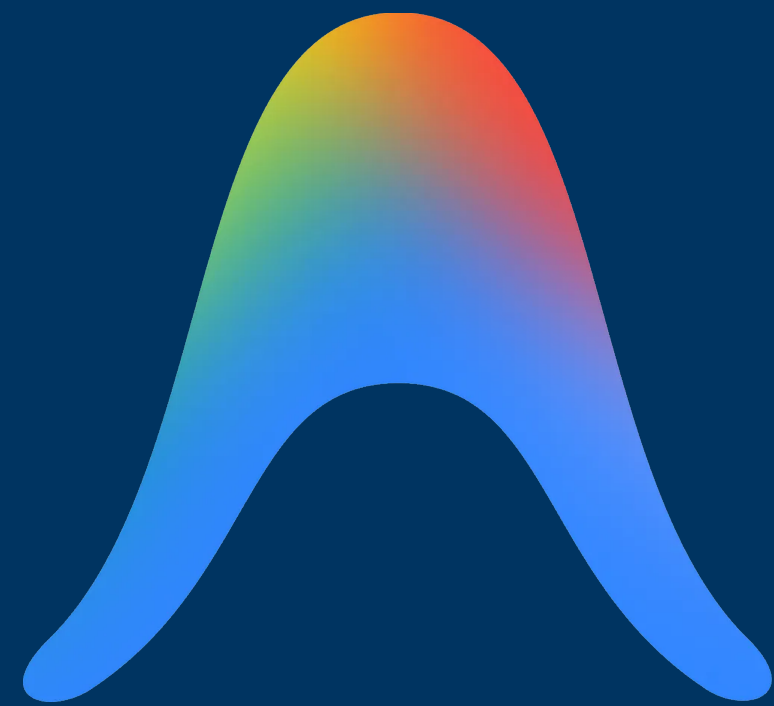
Actions

PR

fork



VS Code



Antigravity



Cursor

*각 아이콘 클릭 시 설치 링크로 연결

AntigravityFileEditSelectionViewGoRunTerminalWindowHelp

2025SDIJ-ASR — train_math.pyOpen Agent ManagerRestart to Update →

Source Control

Changes

Message (⌘Enter to commit on "main")Generate

Commit

Changes3

create_english_adapter.sh251219-sidae-house-f...U

reproduce_tokenizer.py251219-sidae-house-ft/tr...D

train_english.py251219-sidae-house-ft/trainingU

Graph

Auto

OOM RainyForest23main

OOM RainyForest23

Experiment B test RainyForest23

merge issue RainyForest23

merge issue RainyForest23

merge issue RainyForest23

merge issue RainyForest23

merge issue RainyForest23

merge issue RainyForest23

merge issue RainyForest23

adapter test RainyForest23

adapter test RainyForest23

adapter test RainyForest23

adapter test RainyForest23

merge_adapter.pyreproduce_tokenizer.pyD

251219-sidae-house-ft > training > train_math.py

```
474 except Exception as e_local:
475     except Exception as e_hf:
476         metric = _CERMetricFallback()
477
478 def compute_metrics(eval_pred):
479     preds, labels = eval_pred
480     import re
481     import numpy as np
482
483     _CLEAN = re.compile(r"<|>|[\^>]+?>")
484
485     # ★ Replace -100 in labels with pad_token_id
486     labels = np.where(labels != -100, labels, processor.get_token_id("pad"))
487
488     try:
489         pred_str = [_CLEAN.sub("", s).strip() for s in processor.batch_decode(preds)]
490         label_str = [_CLEAN.sub("", s).strip() for s in processor.batch_decode(labels)]
491
492     except Exception as e:
493         logger.error(f"Decode error: {e}")
494         return {"cer": 100.0}
495
496     # 디버깅: 첫 번째 평가에서 샘플 확인
497     if not hasattr(compute_metrics, '_logged'):
498         compute_metrics._logged = True
499         logger.info(f"[COMPUTE METRICS - Sample]")
500         logger.info(f"- Prediction[0]: {pred_str[0]}")
501         logger.info(f"- Reference[0]: {label_str[0]}")
502
503     cer_score = metric.compute(predictions=pred_str, references=label_str)
504
505     # Handle different metric output formats
506     if isinstance(cer_score, dict):
507         cer_score = cer_score.get("cer", 0.0)
508
509     return {"cer": cer_score * 100}
```

Ln 551, Col 20Spaces: 4

Analyzing Training Results

Thought for 15s

학습 결과 및 로그 분석

로그를 바탕으로 분석한 결과, 학습 실패(Loss 폭발 및 생성 오류)의 치명적인 원인이 코드 내에 존재함을 확인했습니다.

1. 원인 분석: 잘못된 <|startoftranscript|> 강제 삽입

로그의 가장 중요한 단서는 DataCollator 디버깅 메시지입니다:

First 10 label tokens (text):
['<|startoftranscript|>', '<|ko|>', '<|transcribe|>', '<|notimestamps|>', ...]

labels(정답 데이터)가 <|startoftranscript|> (SOT)로 시작하고 있습니다. 이는 train_house.py 의 385~392번째 줄에 있는 SOT 강제 삽입 코드 때문입니다.

왜 이것이 문제인가요? Whisper 모델의 학습 과정(Teacher Forcing)은 다음과 같이 작동합니다:

1. 입력(Decoder Input): <|startoftranscript|> (SOT)

2. 예측 목표(Label): 그 다음 나와야 할 토큰 (예: <|ko|>)

하지만 현재 코드는 Label의 맨 앞에도 SOT를

0 Files With ChangesReview Changes

지금까지 내용 깃허브 커밋&푸시 해줘.

PlanningGemini 3 Pro (High)

Antigravity

File

Edit

Selection

View

Go

Run

Terminal

Window

Help

2025SDIJ-ASR — 168fa73 - merge issue (1 file)

Open Agent Manager

Restart to Update →

Source Control

Changes

Message (%Enter to commit on "mai... Generate

Commit

Changes

create_english_adapter.sh 251219-sidae-house-f... U

reproduce_tokenizer.py 251219-sidae-house-ft/tr... D

train_english.py 251219-sidae-house-ft/training U

Graph

Auto

main

OOM RainyForest23

OOM RainyForest23

Experiment B test RainyForest23

merge issue RainyForest23

train_math.py

168fa73 - merge issue (1 file)

00_adapter_merge.py

tmp-test.py

03_run_asr.py

merge_adapter.py

reproduce_tokenizer.py

evaluate_simple.py 251219-sidae-house-ft/pipelines/03_inference

122 hidden lines

123

124 def main():

125 parser = argparse.ArgumentParser(description="Simple evaluation script")

126 parser.add_argument("--model", required=True, help="Path to merged model (.pt)

127 parser.add_argument("--test_manifest", required=True, help="Path to test mani

128 parser.add_argument("--num_samples", type=int, default=15, help="Number of sa

129 parser.add_argument("--device", default="cuda", help="Device (cuda/cpu)")

4 hidden lines

134 # Set seed

135 random.seed(args.seed)

136

137 # Load model

138 model = load_merged_model(args.model, device=args.device)

139

140 # Load test samples

141 print(f"\nLoading {args.num_samples} test samples...")

142 samples = load_test_samples(args.test_manifest, args.num_samples)

143 print(f"✓ Loaded {len(samples)} samples")

144

145 # Evaluate

146 results = transcribe_and_evaluate(model, samples, args.device)

147

148 # Calculate statistics

149 cer_scores = [r['cer'] for r in results]

150 mean_cer = sum(cer_scores) / len(cer_scores)

151

152 print("\n" + "="*60)

153 print("EVALUATION SUMMARY")

154 print("="*60)

155 print(f"CER on {len(samples)} samples: {mean_cer:.2f}%")

156

157 print(f"Min CER: {min(cer_scores):.2f}%")

158

159

160

161

162

163

164

165

166

167

168

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177

122 hidden lines

123

124 def main():

125 parser = argparse.ArgumentParser(description="Simple evaluation script")

126 parser.add_argument("--models", nargs='+', required=True, help="Path(s) to me

127 parser.add_argument("--model_names", nargs='+', help="Names for each model (o

128 parser.add_argument("--test_manifest", required=True, help="Path to test mani

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4 hidden lines

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Irs1_최종.py

Irs1_최최종.py

Irs1_최최최종.py

Irs1_진짜짚막최최최종.py

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unified training RainyForest23

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Antigravity - Settings

Git → 개발자와의 커뮤니케이션

Git → 개발자와의 커뮤니케이션

알기만 해도 여러모로 도움된다!

Q&A → @Rf

👉 제 github 구경하러오세요~

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이 슬라이드에서 사용된 코드

```
git config --global user.name "깃허브아이디"
git config --global user.email "깃허브이메일"
cd Desktop
mkdir 2026-lrs
cd 2026-lrs
git init
git remote add origin "복사한 링크"
git branch -M main
git push -u origin main
git add .
git commit -m "commit message (e.g. fix: 버튼 안눌리는 버그 수정)"
git fetch origin main
git push origin main
git add . && git commit -m "커밋메시지" && git push origin main
```

알고있으면 좋은 코드

```
// git command advanced
git status
git log
git clone [https:// ~~]
git diff [branch#1] [branch#2]
git branch [branch_name]
git checkout [branch_name]

// unix/linux command frequently used
ls /* 현재 디렉토리의 파일 리스트(list) 확인
    -l, -a 등의 option 조합 가능 (e.g. ls -a, ls -l) */
pwd /* 현재 디렉토리 위치 확인 */
rm /* 휴지통 거치지 않고 즉시 삭제(remove) */
```